



US 20250280467A1

(19) **United States**

(12) **Patent Application Publication**
LEE et al.

(10) **Pub. No.: US 2025/0280467 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **METHOD AND DEVICE FOR PERFORMING
SIDELINK COMMUNICATION IN
WIRELESS COMMUNICATION SYSTEM**

Publication Classification

(51) **Int. Cl.**

H04W 76/30 (2018.01)

H04W 24/04 (2009.01)

H04W 74/0808 (2024.01)

H04W 92/18 (2009.01)

(52) **U.S. Cl.**

CPC **H04W 76/30** (2018.02); **H04W 24/04**

(2013.01); **H04W 74/0808** (2013.01); **H04W**

92/18 (2013.01)

(71) Applicant: **LG ELECTRONICS INC.**, Seoul
(KR)

(72) Inventors: **Youngdae LEE**, Seoul (KR); **Hanbyul
SEO**, Seoul (KR); **Seungmin LEE**,
Seoul (KR); **Suckchel YANG**, Seoul
(KR); **Seonwook KIM**, Seoul (KR)

(73) Assignee: **LG ELECTRONICS INC.**, Seoul
(KR)

(21) Appl. No.: **18/859,509**

(22) PCT Filed: **Apr. 26, 2023**

(86) PCT No.: **PCT/KR2023/005703**

§ 371 (c)(1),

(2) Date: **Oct. 23, 2024**

Related U.S. Application Data

(60) Provisional application No. 63/335,197, filed on Apr.
26, 2022.

ABSTRACT

A method and a device for performing communication in a wireless communication system are disclosed. A method for performing communication by a first terminal according to one embodiment of the present disclosure may comprise the steps of: receiving, from a base station, first setting information related to a sidelink (SL)-specific listen before talk (LBT) failure detection timer and second setting information related to an SL-specific LBT failure counter; triggering a consistent SL-specific LBT failure on the basis of the SL-specific failure detection timer and the SL-specific LBT failure counter; releasing an SL connection between second terminals associated with the triggered consistent SL-specific LBT failure; and transmitting, to the base station, information indicating at least one of the triggered consistent SL-specific LBT failure or the SL connection release.

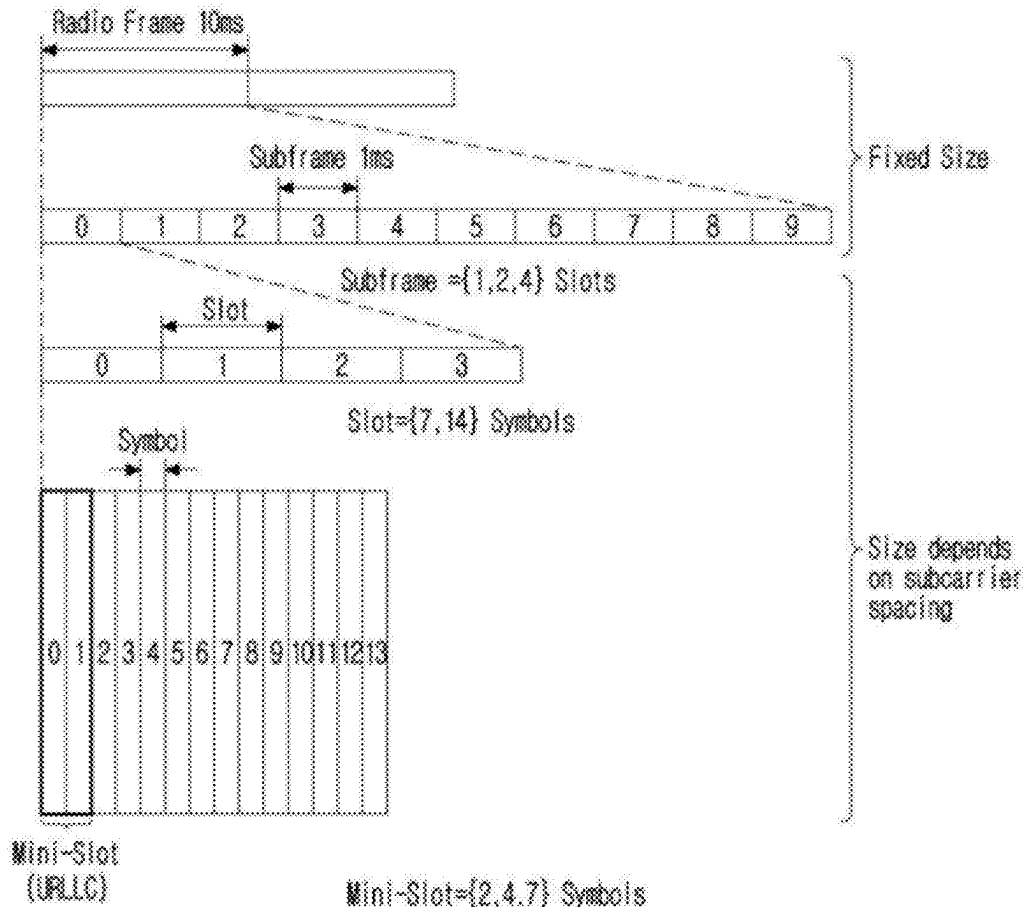


FIG. 1

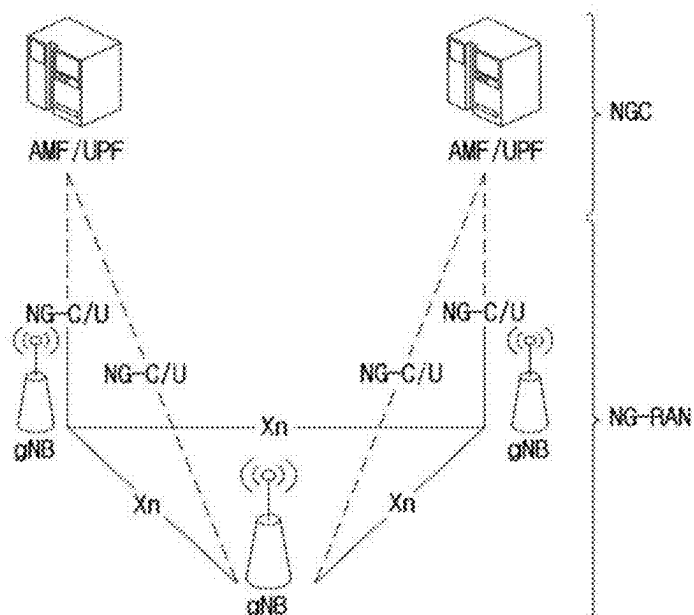


FIG. 2

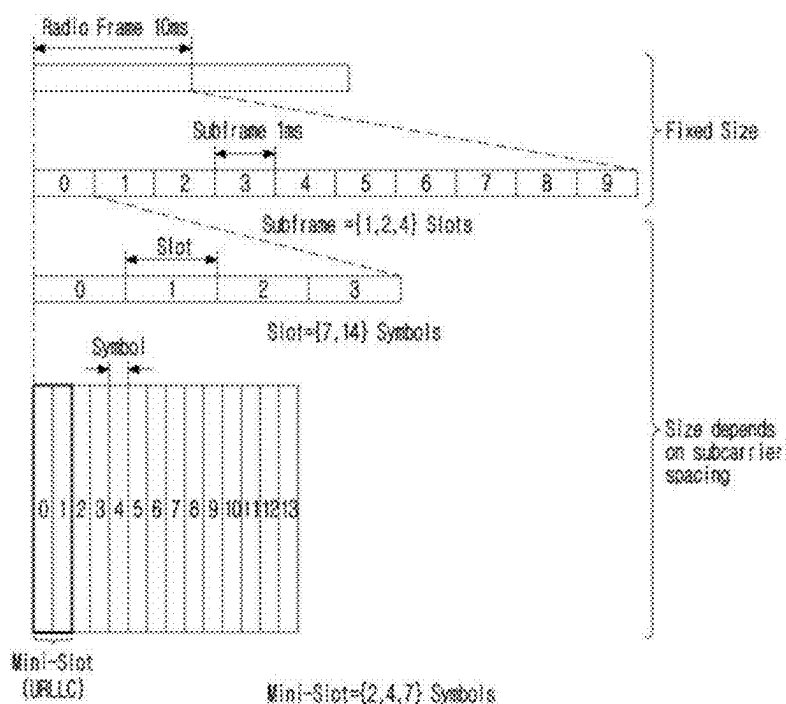


FIG.3

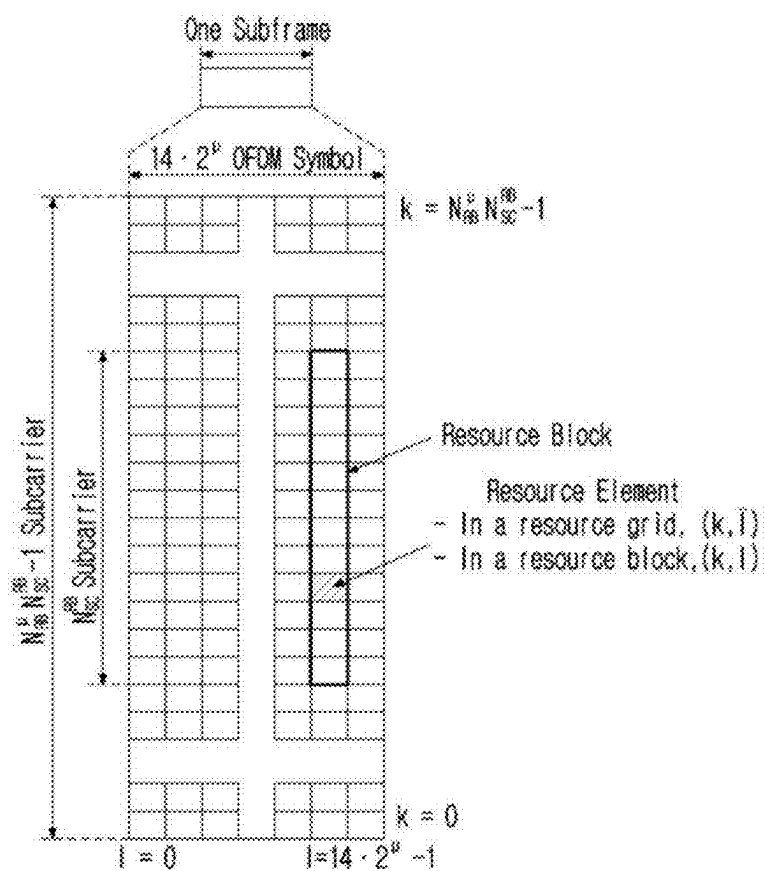


FIG. 4

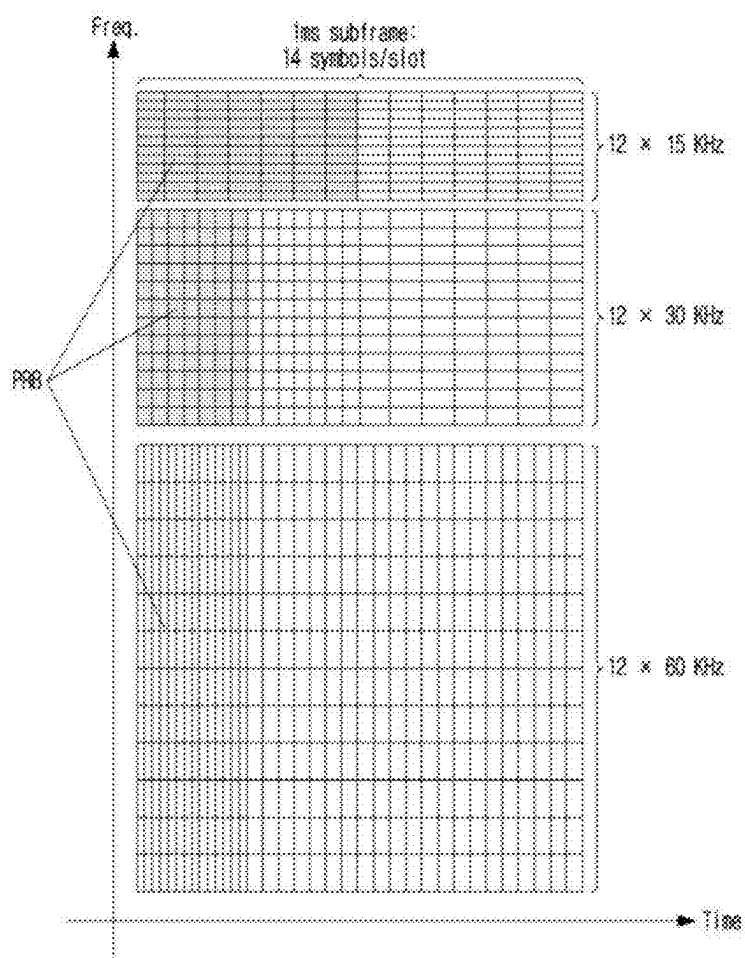


FIG.5

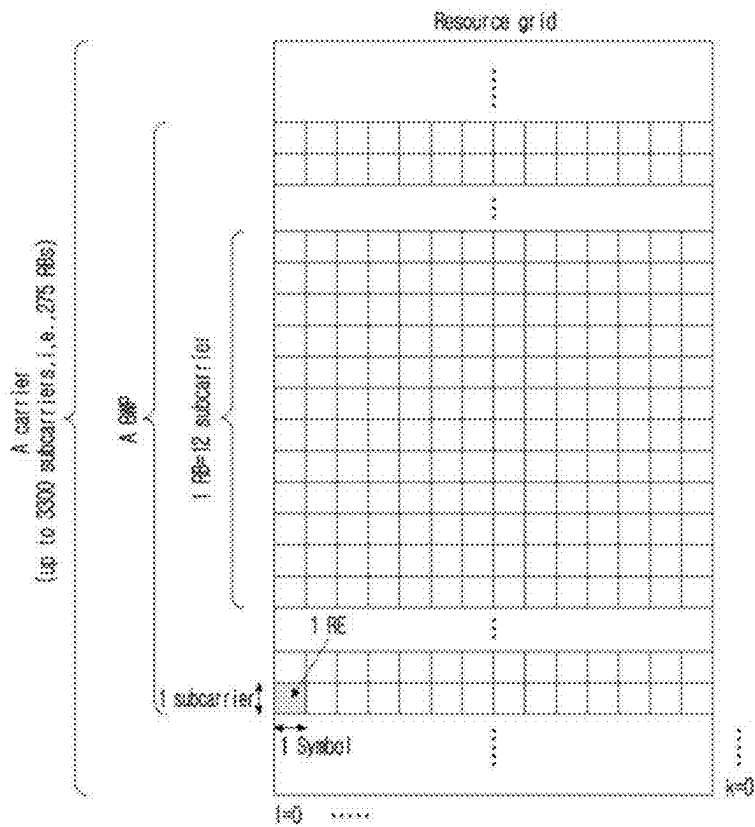


FIG. 6

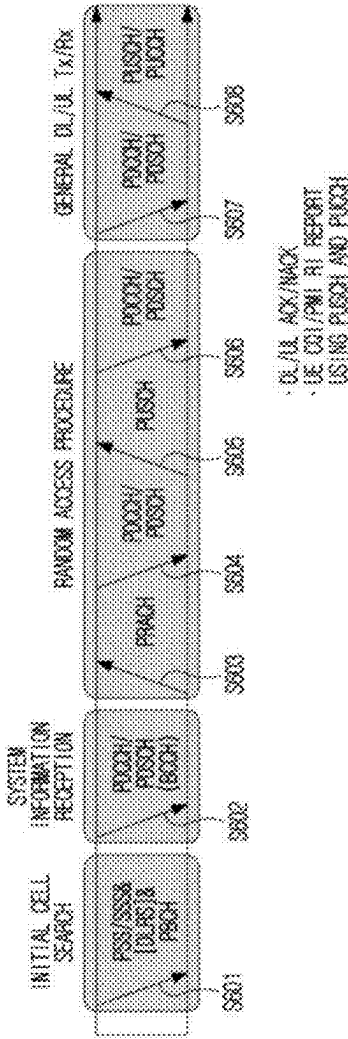


FIG. 7

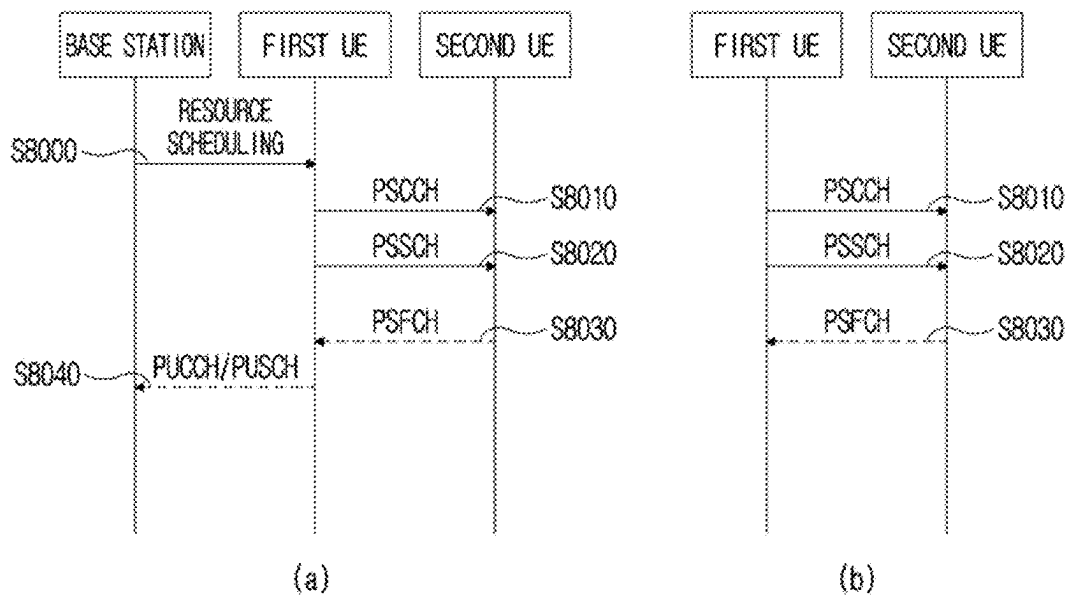


FIG. 8

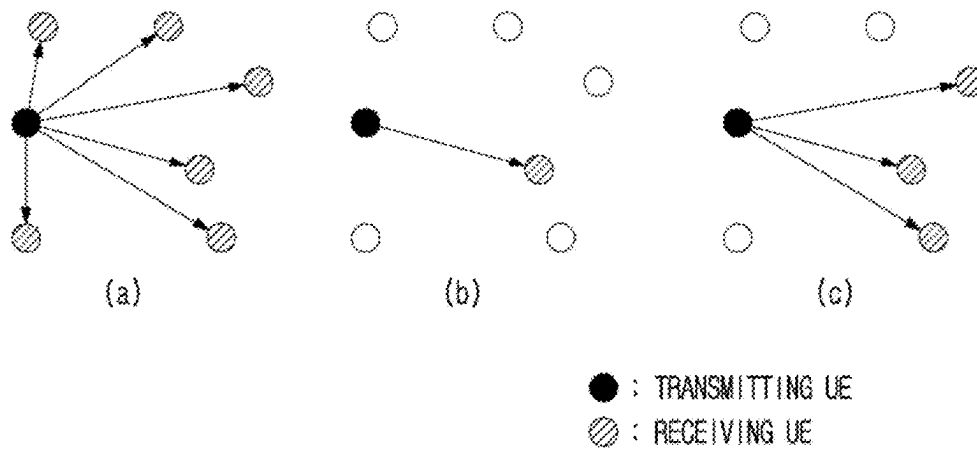


FIG. 9

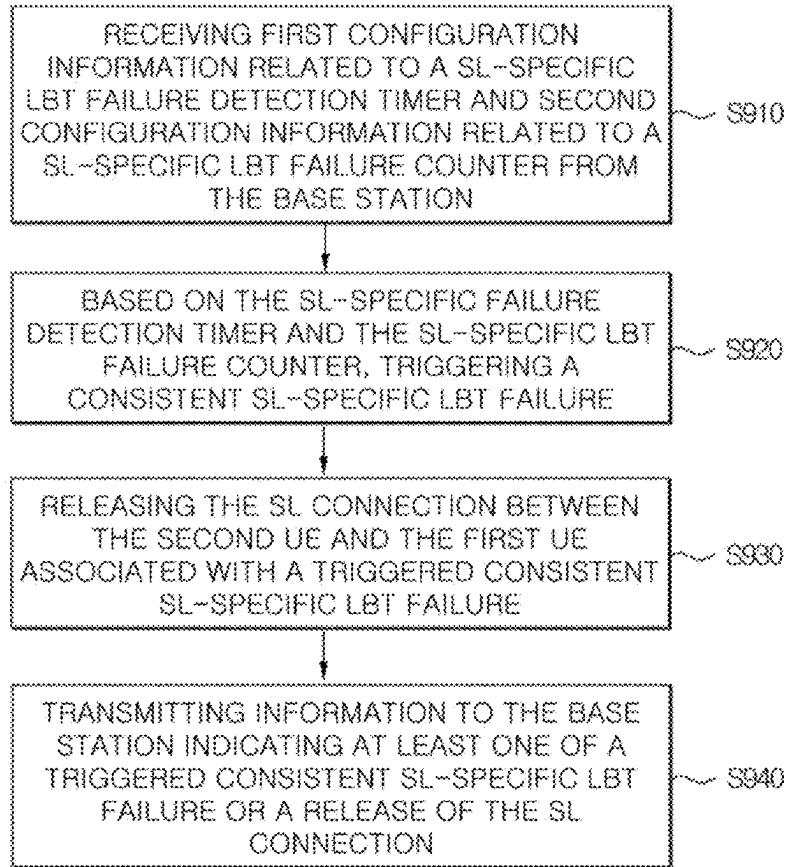


FIG. 10

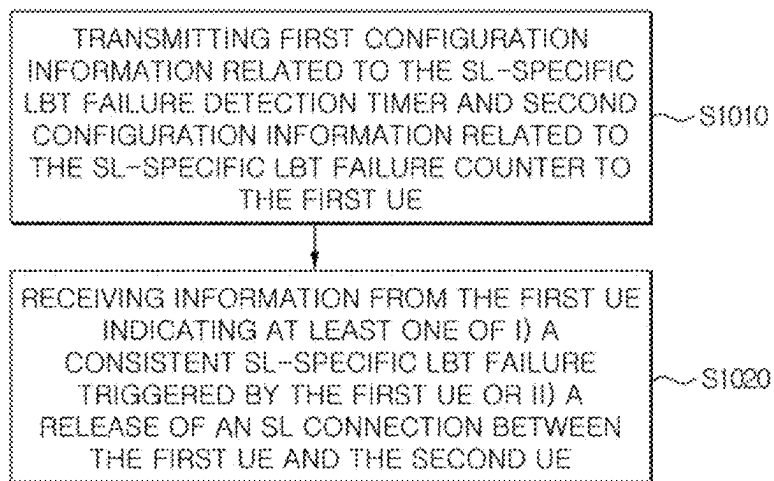


FIG. 11

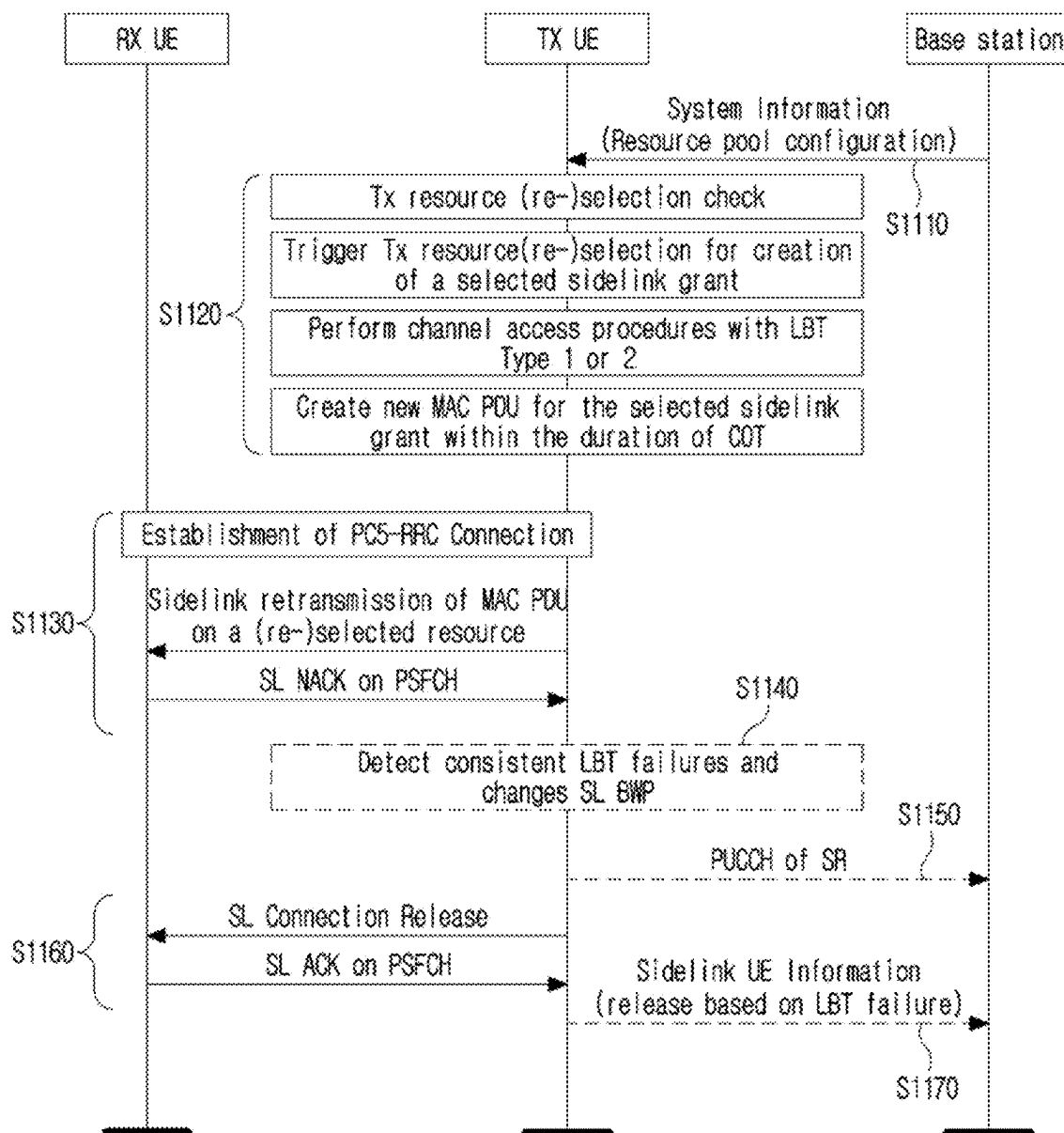


FIG. 12

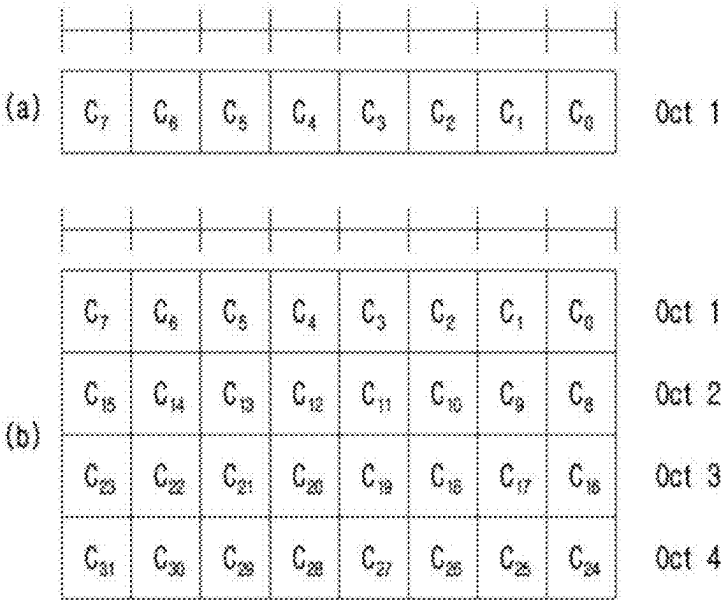
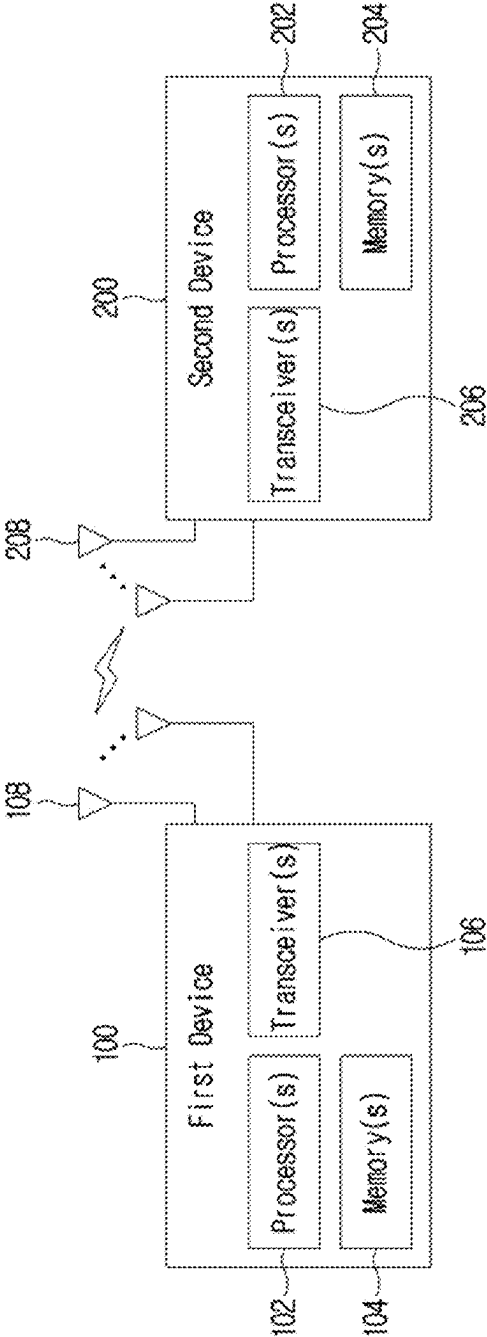


FIG. 13



METHOD AND DEVICE FOR PERFORMING SIDELINK COMMUNICATION IN WIRELESS COMMUNICATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2023/005703, filed on Apr. 26, 2023, which claims the benefit of U.S. Provisional Application No. 63/335,197, filed on Apr. 26, 2022, the contents of which are all incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a wireless communication system, and more particularly, to a method and device for performing sidelink communication in a wireless communication system.

BACKGROUND

[0003] A mobile communication system has been developed to provide a voice service while guaranteeing mobility of users. However, a mobile communication system has extended even to a data service as well as a voice service, and currently, an explosive traffic increase has caused shortage of resources and users have demanded a faster service, so a more advanced mobile communication system has been required.

[0004] The requirements of a next-generation mobile communication system at large should be able to support accommodation of explosive data traffic, a remarkable increase in a transmission rate per user, accommodation of the significantly increased number of connected devices, very low End-to-End latency and high energy efficiency. To this end, a variety of technologies such as Dual Connectivity, Massive Multiple Input Multiple Output (Massive MIMO), In-band Full Duplex, Non-Orthogonal Multiple Access (NOMA), Super wideband Support, Device Networking, etc. have been researched.

SUMMARY

[0005] The technical problem of the present disclosure is to provide a method and device for performing sidelink communication in a wireless communication system.

[0006] The technical problem of the present disclosure is to provide a method and device for detecting/triggering SL-specific LBT failure based on a SL-specific LBT failure related counter and a SL-specific LBT failure detection timer.

[0007] The technical problem of the present disclosure is to provide a method and device for releasing an SL connection related to a triggered/detected SL-specific LBT failure.

[0008] The technical problems to be achieved by the present disclosure are not limited to the above-described technical objects, and other technical problems which are not described herein will be clearly understood by those skilled in the pertinent art from the following description.

[0009] According to one embodiment of the present disclosure, a method performed by a first user equipment (UE) in a wireless communication system may include receiving, from a base station, first configuration information related to a sidelink (SL)-specific LBT (listen before talk) failure detection timer and second configuration information related

to an SL-specific LBT failure counter; triggering a consistent SL-specific LBT failure based on the SL-specific failure detection timer and the SL-specific LBT failure counter; releasing an SL connection of a second UE related to the triggered consistent SL-specific LBT failure; and transmitting, to the base station, information indicating at least one of the triggered consistent SL-specific LBT failure or the release of the SL connection.

[0010] According to one embodiment of the present disclosure, a method performed by a base station in a wireless communication system may include transmitting, to a user equipment (UE), first configuration information related to a sidelink (SL)-specific LBT (listen before talk) failure detection timer and second configuration information related to an SL-specific LBT failure counter; and receiving, from the first UE, information indicating at least one of a consistent SL-specific LBT failure triggered by the first UE or a release of an SL connection between the first UE and the second UE, and the consistent SL-specific LBT failure may be triggered based on the SL-specific failure detection timer and the SL-specific LBT failure counter, and a SL connection between the first UE and the second UE related to the triggered consistent SL-specific LBT failure may be released.

[0011] According to one embodiment of the present disclosure, a method and device for performing sidelink communication in a wireless communication system may be provided.

[0012] According to various embodiments of the present disclosure, a method and device for detecting/triggering SL-specific LBT failure based on a SL-specific LBT failure related counter and a SL-specific LBT failure detection timer may be provided.

[0013] According to various embodiments of the present disclosure, a method and device for releasing and reporting an SL connection related to a triggered/detected SL-specific LBT failure may be provided.

[0014] According to one embodiment of the present disclosure, by receiving information reporting whether an SL connection between terminals is released, a base station can more efficiently operate sidelink resources.

[0015] Effects achievable by the present disclosure are not limited to the above-described effects, and other effects which are not described herein may be clearly understood by those skilled in the pertinent art from the following description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] Accompanying drawings included as part of detailed description for understanding the present disclosure provide embodiments of the present disclosure and describe technical features of the present disclosure with detailed description.

[0017] FIG. 1 illustrates a structure of a wireless communication system to which the present disclosure may be applied.

[0018] FIG. 2 illustrates a frame structure in a wireless communication system to which the present disclosure may be applied.

[0019] FIG. 3 illustrates a resource grid in a wireless communication system to which the present disclosure may be applied.

[0020] FIG. 4 illustrates a physical resource block in a wireless communication system to which the present disclosure may be applied.

[0021] FIG. 5 illustrates a slot structure in a wireless communication system to which the present disclosure may be applied.

[0022] FIG. 6 illustrates physical channels used in a wireless communication system to which the present disclosure may be applied and a general signal transmission and reception method using them.

[0023] FIG. 7 illustrates a procedure for performing V2X or SL communication according to a transmission mode in a wireless communication system to which the present disclosure can be applied.

[0024] FIG. 8 illustrates a cast type for V2X or SL communication in a wireless communication system to which the present disclosure may be applied.

[0025] FIG. 9 is a diagram for explaining a process in which a first UE performs communication according to an embodiment of the present disclosure.

[0026] FIG. 10 is a diagram for explaining a process in which a base station performs communication according to an embodiment of the present disclosure.

[0027] FIG. 11 is a diagram for explaining a signaling procedure of a base station, a first UE, and a second UE according to an embodiment of the present disclosure.

[0028] FIG. 12 is a diagram for explaining a (SL) LBT failure MAC CE structure according to an embodiment of the present disclosure.

[0029] FIG. 13 illustrates a block diagram of a wireless communication device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0030] Hereinafter, embodiments according to the present disclosure will be described in detail by referring to accompanying drawings. Detailed description to be disclosed with accompanying drawings is to describe exemplary embodiments of the present disclosure and is not to represent the only embodiment that the present disclosure may be implemented. The following detailed description includes specific details to provide complete understanding of the present disclosure. However, those skilled in the pertinent art knows that the present disclosure may be implemented without such specific details.

[0031] In some cases, known structures and devices may be omitted or may be shown in a form of a block diagram based on a core function of each structure and device in order to prevent a concept of the present disclosure from being ambiguous.

[0032] In the present disclosure, when an element is referred to as being “connected”, “combined” or “linked” to another element, it may include an indirect connection relation that yet another element presents therebetween as well as a direct connection relation. In addition, in the present disclosure, a term, “include” or “have”, specifies the presence of a mentioned feature, step, operation, component and/or element, but it does not exclude the presence or addition of one or more other features, stages, operations, components, elements and/or their groups.

[0033] In the present disclosure, a term such as “first”, “second”, etc. is used only to distinguish one element from other element and is not used to limit elements, and unless otherwise specified, it does not limit an order or importance,

etc. between elements. Accordingly, within a scope of the present disclosure, a first element in an embodiment may be referred to as a second element in another embodiment and likewise, a second element in an embodiment may be referred to as a first element in another embodiment.

[0034] A term used in the present disclosure is to describe a specific embodiment, and is not to limit a claim. As used in a described and attached claim of an embodiment, a singular form is intended to include a plural form, unless the context clearly indicates otherwise. A term used in the present disclosure, “and/or”, may refer to one of related enumerated items or it means that it refers to and includes any and all possible combinations of two or more of them. In addition, “/” between words in the present disclosure has the same meaning as “and/or”, unless otherwise described.

[0035] The present disclosure describes a wireless communication network or a wireless communication system, and an operation performed in a wireless communication network may be performed in a process in which a device (e.g., a base station) controlling a corresponding wireless communication network controls a network and transmits or receives a signal, or may be performed in a process in which a terminal associated to a corresponding wireless network transmits or receives a signal with a network or between terminals.

[0036] In the present disclosure, transmitting or receiving a channel includes a meaning of transmitting or receiving information or a signal through a corresponding channel. For example, transmitting a control channel means that control information or a control signal is transmitted through a control channel. Similarly, transmitting a data channel means that data information or a data signal is transmitted through a data channel.

[0037] Hereinafter, a downlink (DL) means a communication from a base station to a terminal and an uplink (UL) means a communication from a terminal to a base station. In a downlink, a transmitter may be part of a base station and a receiver may be part of a terminal. In an uplink, a transmitter may be part of a terminal and a receiver may be part of a base station. A base station may be expressed as a first communication device and a terminal may be expressed as a second communication device. A base station (BS) may be substituted with a term such as a fixed station, a Node B, an eNB (evolved-NodeB), a gNB (Next Generation NodeB), a BTS (base transceiver system), an Access Point (AP), a Network (5G network), an AI (Artificial Intelligence) system/module, an RSU (road side unit), a robot, a drone (UAV: Unmanned Aerial Vehicle), an AR (Augmented Reality) device, a VR (Virtual Reality) device, etc. In addition, a terminal may be fixed or mobile, and may be substituted with a term such as a UE (User Equipment), an MS (Mobile Station), a UT (user terminal), an MSS (Mobile Subscriber Station), an SS (Subscriber Station), an AMS (Advanced Mobile Station), a WT (Wireless terminal), an MTC (Machine-Type Communication) device, an M2M (Machine-to-Machine) device, a D2D (Device-to-Device) device, a vehicle, an RSU (road side unit), a robot, an AI (Artificial Intelligence) module, a drone (UAV: Unmanned Aerial Vehicle), an AR (Augmented Reality) device, a VR (Virtual Reality) device, etc.

[0038] The following description may be used for a variety of radio access systems such as CDMA, FDMA, TDMA, OFDMA, SC-FDMA, etc. CDMA may be implemented by a wireless technology such as UTRA (Universal Terrestrial

Radio Access) or CDMA2000. TDMA may be implemented by a radio technology such as GSM (Global System for Mobile communications)/GPRS (General Packet Radio Service)/EDGE (Enhanced Data Rates for GSM Evolution). OFDMA may be implemented by a radio technology such as IEEE 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802-20, E-UTRA (Evolved UTRA), etc. UTRA is a part of a UMTS (Universal Mobile Telecommunications System). 3GPP (3rd Generation Partnership Project) LTE (Long Term Evolution) is a part of an E-UMTS (Evolved UMTS) using E-UTRA and LTE-A (Advanced)/LTE-A pro is an advanced version of 3GPP LTE. 3GPP NR (New Radio or New Radio Access Technology) is an advanced version of 3GPP LTE/LTE-A/LTE-A pro.

[0039] To clarify description, it is described based on a 3GPP communication system (e.g., LTE-A, NR), but a technical idea of the present disclosure is not limited thereto. LTE means a technology after 3GPP TS (Technical Specification) 36.xxx Release 8. In detail, an LTE technology in or after 3GPP TS 36.xxx Release 10 is referred to as LTE-A and an LTE technology in or after 3GPP TS 36.xxx Release 13 is referred to as LTE-A pro. 3GPP NR means a technology in or after TS 38.xxx Release 15. LTE/NR may be referred to as a 3GPP system. “xxx” means a detailed number for a standard document. LTE/NR may be commonly referred to as a 3GPP system. For a background art, a term, an abbreviation, etc. used to describe the present disclosure, matters described in a standard document disclosed before the present disclosure may be referred to. For example, the following document may be referred to.

[0040] For 3GPP LTE, TS 36.211 (physical channels and modulation), TS 36.212 (multiplexing and channel coding), TS 36.213 (physical layer procedures), TS 36.300 (overall description), TS 36.331 (radio resource control) may be referred to.

[0041] For 3GPP NR, TS 38.211 (physical channels and modulation), TS 38.212 (multiplexing and channel coding), TS 38.213 (physical layer procedures for control), TS 38.214 (physical layer procedures for data), TS 38.300 (NR and NG-RAN (New Generation-Radio Access Network) overall description), TS 38.331 (radio resource control protocol specification) may be referred to.

[0042] Abbreviations of terms which may be used in the present disclosure is defined as follows.

- [0043] BM: beam management
- [0044] CQI: Channel Quality Indicator
- [0045] CRI: channel state information-reference signal resource indicator
- [0046] CSI: channel state information
- [0047] CSI-IM: channel state information-interference measurement
- [0048] CSI-RS: channel state information-reference signal
- [0049] DMRS: demodulation reference signal
- [0050] FDM: frequency division multiplexing
- [0051] FFT: fast Fourier transform
- [0052] IFDMA: interleaved frequency division multiple access
- [0053] IFFT: inverse fast Fourier transform
- [0054] L1-RSRP: Layer 1 reference signal received power
- [0055] L1-RSRQ: Layer 1 reference signal received quality
- [0056] MAC: medium access control

- [0057] NZP: non-zero power
- [0058] OFDM: orthogonal frequency division multiplexing
- [0059] PDCCH: physical downlink control channel
- [0060] PDSCH: physical downlink shared channel
- [0061] PMI: precoding matrix indicator
- [0062] RE: resource element
- [0063] RI: Rank indicator
- [0064] RRC: radio resource control
- [0065] RSSI: received signal strength indicator
- [0066] Rx: Reception
- [0067] QCL: quasi co-location
- [0068] SINR: signal to interference and noise ratio
- [0069] SSB (or SS/PBCH block): Synchronization signal block (including PSS (primary synchronization signal), SSS (secondary synchronization signal) and PBCH (physical broadcast channel))
- [0070] TDM: time division multiplexing
- [0071] TRP: transmission and reception point
- [0072] TRS: tracking reference signal
- [0073] Tx: transmission
- [0074] UE: user equipment
- [0075] ZP: zero power

Overall System

[0076] As more communication devices have required a higher capacity, a need for an improved mobile broadband communication compared to the existing radio access technology (RAT) has emerged. In addition, massive MTC (Machine Type Communications) providing a variety of services anytime and anywhere by connecting a plurality of devices and things is also one of main issues which will be considered in a next-generation communication. Furthermore, a communication system design considering a service/a terminal sensitive to reliability and latency is also discussed. As such, introduction of a next-generation RAT considering eMBB (enhanced mobile broadband communication), mMTC (massive MTC), URLLC (Ultra-Reliable and Low Latency Communication), etc. is discussed and, for convenience, a corresponding technology is referred to as NR in the present disclosure. NR is an expression which represents an example of a 5G RAT.

[0077] A new RAT system including NR uses an OFDM transmission method or a transmission method similar to it. A new RAT system may follow OFDM parameters different from OFDM parameters of LTE. Alternatively, a new RAT system follows a numerology of the existing LTE/LTE-A as it is, but may support a wider system bandwidth (e.g., 100 MHz). Alternatively, one cell may support a plurality of numerologies. In other words, terminals which operate in accordance with different numerologies may coexist in one cell.

[0078] A numerology corresponds to one subcarrier spacing in a frequency domain. As a reference subcarrier spacing is scaled by an integer N, a different numerology may be defined.

[0079] FIG. 1 illustrates a structure of a wireless communication system to which the present disclosure may be applied.

[0080] In reference to FIG. 1, NG-RAN is configured with gNBs which provide a control plane (RRC) protocol end for a NG-RAN (NG-Radio Access) user plane (i.e., a new AS (access stratum) sublayer/PDCP (Packet Data Convergence Protocol)/RLC (Radio Link Control)/MAC/PHY) and UE.

The gNBs are interconnected through a Xn interface. The gNB, in addition, is connected to an NGC (New Generation Core) through an NG interface. In more detail, the gNB is connected to an AMF (Access and Mobility Management Function) through an N2 interface, and is connected to a UPF (User Plane Function) through an N3 interface.

[0081] FIG. 2 illustrates a frame structure in a wireless communication system to which the present disclosure may be applied.

[0082] A NR system may support a plurality of numerologies. Here, a numerology may be defined by a subcarrier spacing and a cyclic prefix (CP) overhead. Here, a plurality of subcarrier spacings may be derived by scaling a basic (reference) subcarrier spacing by an integer N (or, u). In addition, although it is assumed that a very low subcarrier spacing is not used in a very high carrier frequency, a used numerology may be selected independently from a frequency band. In addition, a variety of frame structures according to a plurality of numerologies may be supported in a NR system.

[0083] Hereinafter, an OFDM numerology and frame structure which may be considered in a NR system will be described. A plurality of OFDM numerologies supported in a NR system may be defined as in the following Table 1.

TABLE 1

μ	$\Delta f = 2^\mu \cdot 15$ [kHz]	CP
0	15	Normal
1	30	Normal
2	60	Normal, Extended
3	120	Normal
4	240	Normal

[0084] NR supports a plurality of numerologies (or subcarrier spacings (SCS)) for supporting a variety of 5G services. For example, when a SCS is 15 kHz, a wide area in traditional cellular bands is supported, and when a SCS is 30 KHz/60 kHz, dense-urban, lower latency and a wider carrier bandwidth are supported, and when a SCS is 60 kHz or higher, a bandwidth wider than 24.25 GHz is supported to overcome a phase noise. An NR frequency band is defined as a frequency range in two types (FR1, FR2). FR1, FR2 may be configured as in the following Table 2. In addition, FR2 may mean a millimeter wave (mmW).

TABLE 2

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing
FR1	410 MHz-7125 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

[0085] Regarding a frame structure in an NR system, a size of a variety of fields in a time domain is expressed as a multiple of a time unit of $T_c = 1/(\Delta f_{max} \cdot N_f)$. Here, Δf_{max} is $480 \cdot 10^3$ Hz and N_f is 4096. Downlink and uplink transmission is configured (organized) with a radio frame having a duration of $T_f = 1/(\Delta f_{max} \cdot N_f / 100) \cdot T_c = 10$ ms. Here, a radio frame is configured with 10 subframes having a duration of $T_{sf} = (\Delta f_{max} \cdot N_f / 1000) \cdot T_c = 1$ ms, respectively. In this case, there may be one set of frames for an uplink and one set of frames for a downlink. In addition, transmission in an uplink frame No. i from a terminal should start earlier by $T_{TA} = (N_{TA} + N_{TA, offset})T_c$ than a corresponding downlink frame in a corresponding terminal starts. For a subcarrier spacing configuration μ , slots are numbered in an increasing order of $n_s^\mu \in \{0, \dots, N_{slot}^{subframe, \mu} - 1\}$ in a subframe and are numbered in an increasing order of $n_{sf}^\mu \in \{0, \dots, N_{slot}^{frame, \mu} - 1\}$ in a radio frame. One slot is configured with N_{symb}^{slot} consecutive OFDM symbols and N_{symb}^{slot} is determined according to CP. A start of a slot n_s^μ in a subframe is temporally arranged with a start of an OFDM symbol $n_s^\mu N_{symb}^{slot}$ in the same subframe. All terminals may not perform transmission and reception at the same time, which means that all OFDM symbols of a downlink slot or an uplink slot may not be used.

[0086] Table 3 represents the number of OFDM symbols per slot (N_{symb}^{slot}), the number of slots per radio frame ($N_{slot}^{frame, \mu}$) and the number of slots per subframe ($N_{slot}^{subframe, \mu}$) in a normal CP and Table 4 represents the number of OFDM symbols per slot, the number of slots per radio frame and the number of slots per subframe in an extended CP.

TABLE 3

μ	N_{symb}^{slot}	$N_{slot}^{frame, \mu}$	$N_{slot}^{subframe, \mu}$
0	14	10	1
1	14	20	2
2	14	40	4
3	14	80	8
4	14	160	16

TABLE 4

μ	N_{symb}^{slot}	$N_{slot}^{frame, \mu}$	$N_{slot}^{subframe, \mu}$
2	12	40	4

[0087] FIG. 2 is an example on $\mu=2$ (SCS is 60 kHz), 1 subframe may include 4 slots referring to Table 3. 1 subframe={1,2,4} slot shown in FIG. 2 is an example, the number of slots which may be included in 1 subframe is defined as in Table 3 or Table 4. In addition, a mini-slot may include 2, 4 or 7 symbols or more or less symbols. Regarding a physical resource in a NR system, an antenna port, a resource grid, a resource element, a resource block, a carrier part, etc. may be considered. Hereinafter, the physical resources which may be considered in an NR system will be described in detail.

[0088] First, in relation to an antenna port, an antenna port is defined so that a channel where a symbol in an antenna port is carried can be inferred from a channel where other symbol in the same antenna port is carried. When a large-scale property of a channel where a symbol in one antenna port is carried may be inferred from a channel where a symbol in other antenna port is carried, it may be said that 2 antenna ports are in a QC/QCL (quasi co-located or quasi co-location) relationship. In this case, the large-scale property includes at least one of delay spread, doppler spread, frequency shift, average received power, received timing.

[0089] FIG. 3 illustrates a resource grid in a wireless communication system to which the present disclosure may be applied.

[0090] In reference to FIG. 3, it is illustratively described that a resource grid is configured with $N_{RB}^\mu N_{sc}^{RB}$ subcarriers in a frequency domain and one subframe is configured with $14 \cdot 2^\mu$ OFDM symbols, but it is not limited thereto. In an NR

system, a transmitted signal is described by OFDM symbols of $2^{\mu}N_{sc}^{symb(\mu)}$ and one or more resource grids configured with $N_{RB}^{\mu}N_{sc}^{RB}$ subcarriers. Here, $N_{RB}^{\mu} \leq N_{RB}^{max,\mu}$. The $N_{RB}^{max,\mu}$ represents a maximum transmission bandwidth, which may be different between an uplink and a downlink as well as between numerologies. In this case, one resource grid may be configured per μ and an antenna port p . Each element of a resource grid for μ and an antenna port p is referred to as a resource element and is uniquely identified by an index pair (k,l') . Here, $k=0, \dots, N_{RB}^{\mu}N_{sc}^{RB}-1$ is an index in a frequency domain and $l'=0, \dots, 2^{\mu}N_{sc}^{symb(\mu)}-1$ refers to a position of a symbol in a subframe. When referring to a resource element in a slot, an index pair (k,l) is used.

[0091] Here, $l=0, \dots, N_{sc}^{symb}-1$. A resource element (k,l') for μ and an antenna port p corresponds to a complex value, $a_{k,l'}^{(p,\mu)}$. When there is no risk of confusion or when a specific antenna port or numerology is not specified, indexes p and μ may be dropped, whereupon a complex value may be $a_{k,l'}^{(p)}$ or $a_{k,l'}$. In addition, a resource block (RB) is defined as $N_{sc}^{RB}=12$ consecutive subcarriers in a frequency domain.

[0092] Point A plays a role as a common reference point of a resource block grid and is obtained as follows.

[0093] `offsetToPointA` for a primary cell (PCell) downlink represents a frequency offset between point A and the lowest subcarrier of the lowest resource block overlapped with a SS/PBCH block which is used by a terminal for an initial cell selection. It is expressed in resource block units assuming a 15 kHz subcarrier spacing for FR1 and a 60 kHz subcarrier spacing for FR2.

[0094] `absoluteFrequency PointA` represents a frequency-position of point A expressed as in ARFCN (absolute radio-frequency channel number).

[0095] Common resource blocks are numbered from 0 to the top in a frequency domain for a subcarrier spacing configuration μ . The center of subcarrier 0 of common resource block 0 for a subcarrier spacing configuration μ is identical to 'point A'. A relationship between a common resource block number n_{CRB}^{μ} and a resource element (k,l) for a subcarrier spacing configuration μ in a frequency domain is given as in the following Equation 1.

$$n_{CRB}^{\mu} = \left\lfloor \frac{k}{N_{sc}^{RB}} \right\rfloor \quad [\text{Equation 1}]$$

[0096] In Equation 1, k is defined relatively to point A so that $k=0$ corresponds to a subcarrier centering in point A. Physical resource blocks are numbered from 0 to $N_{BWP,i}^{size}-1$ in a bandwidth part (BWP) and i is a number of a BWP. A relationship between a physical resource block n_{PRB} and a common resource block n_{CRB} in BWP i is given by the following Equation 2.

$$n_{CRB}^{\mu} = n_{PRB}^{\mu} + N_{BWP,i}^{start,\mu} \quad [\text{Equation 2}]$$

[0097] $N_{BWP,i}^{start,\mu}$ is a common resource block that a BWP starts relatively to common resource block 0.

[0098] FIG. 4 illustrates a physical resource block in a wireless communication system to which the present disclosure may be applied. And, FIG. 5 illustrates a slot

structure in a wireless communication system to which the present disclosure may be applied.

[0099] In reference to FIG. 4 and FIG. 5, a slot includes a plurality of symbols in a time domain. For example, for a normal CP, one slot includes 7 symbols, but for an extended CP, one slot includes 6 symbols.

[0100] A carrier includes a plurality of subcarriers in a frequency domain. An RB (Resource Block) is defined as a plurality of (e.g., 12) consecutive subcarriers in a frequency domain. A BWP (Bandwidth Part) is defined as a plurality of consecutive (physical) resource blocks in a frequency domain and may correspond to one numerology (e.g., an SCS, a CP length, etc.). A carrier may include a maximum N (e.g., 5) BWPs. A data communication may be performed through an activated BWP and only one BWP may be activated for one terminal. In a resource grid, each element is referred to as a resource element (RE) and one complex symbol may be mapped.

[0101] In an NR system, up to 400 MHz may be supported per component carrier (CC). If a terminal operating in such a wideband CC always operates turning on a radio frequency (FR) chip for the whole CC, terminal battery consumption may increase. Alternatively, when several application cases operating in one wideband CC (e.g., eMBB, URLLC, Mmtc, V2X, etc.) are considered, a different numerology (e.g., a subcarrier spacing, etc.) may be supported per frequency band in a corresponding CC. Alternatively, each terminal may have a different capability for the maximum bandwidth. By considering it, a base station may indicate a terminal to operate only in a partial bandwidth, not in a full bandwidth of a wideband CC, and a corresponding partial bandwidth is defined as a bandwidth part (BWP) for convenience. A BWP may be configured with consecutive RBs on a frequency axis and may correspond to one numerology (e.g., a subcarrier spacing, a CP length, a slot/a mini-slot duration).

[0102] Meanwhile, a base station may configure a plurality of BWPs even in one CC configured to a terminal. For example, a BWP occupying a relatively small frequency domain may be configured in a PDCCH monitoring slot, and a PDSCH indicated by a PDCCH may be scheduled in a greater BWP. Alternatively, when UEs are congested in a specific BWP, some terminals may be configured with other BWP for load balancing. Alternatively, considering frequency domain inter-cell interference cancellation between neighboring cells, etc., some middle spectrums of a full bandwidth may be excluded and BWPs on both edges may be configured in the same slot. In other words, a base station may configure at least one DL/UL BWP to a terminal associated with a wideband CC. A base station may activate at least one DL/UL BWP of configured DL/UL BWP(s) at a specific time (by L1 signaling or MAC CE (Control Element) or RRC signaling, etc.).

[0103] In addition, a base station may indicate switching to other configured DL/UL BWP (by L1 signaling or MAC CE or RRC signaling, etc.). Alternatively, based on a timer, when a timer value is expired, it may be switched to a determined DL/UL BWP. Here, an activated DL/UL BWP is defined as an active DL/UL BWP. But, a configuration on a DL/UL BWP may not be received when a terminal performs an initial access procedure or before a RRC connection is set up, so a DL/UL BWP which is assumed by a terminal under these situations is defined as an initial active DL/UL BWP.

[0104] FIG. 6 illustrates physical channels used in a wireless communication system to which the present disclosure

may be applied and a general signal transmission and reception method using them.

[0105] In a wireless communication system, a terminal receives information through a downlink from a base station and transmits information through an uplink to a base station. Information transmitted and received by a base station and a terminal includes data and a variety of control information and a variety of physical channels exist according to a type/a usage of information transmitted and received by them.

[0106] When a terminal is turned on or newly enters a cell, it performs an initial cell search including synchronization with a base station or the like (S601). For the initial cell search, a terminal may synchronize with a base station by receiving a primary synchronization signal (PSS) and a secondary synchronization signal (SSS) from a base station and obtain information such as a cell identifier (ID), etc. After that, a terminal may obtain broadcasting information in a cell by receiving a physical broadcast channel (PBCH) from a base station. Meanwhile, a terminal may check out a downlink channel state by receiving a downlink reference signal (DL RS) at an initial cell search stage.

[0107] A terminal which completed an initial cell search may obtain more detailed system information by receiving a physical downlink control channel (PDCCH) and a physical downlink shared channel (PDSCH) according to information carried in the PDCCH (S602).

[0108] Meanwhile, when a terminal accesses to a base station for the first time or does not have a radio resource for signal transmission, it may perform a random access (RACH) procedure to a base station (S603 to S606). For the random access procedure, a terminal may transmit a specific sequence as a preamble through a physical random access channel (PRACH) (S603 and S605) and may receive a response message for a preamble through a PDCCH and a corresponding PDSCH (S604 and S606). A contention based RACH may additionally perform a contention resolution procedure.

[0109] A terminal which performed the above-described procedure subsequently may perform PDCCH/PDSCH reception (S607) and PUSCH (Physical Uplink Shared Channel)/PUCCH (physical uplink control channel) transmission (S608) as a general uplink/downlink signal transmission procedure. In particular, a terminal receives downlink control information (DCI) through a PDCCH. Here, DCI includes control information such as resource allocation information for a terminal and a format varies depending on its purpose of use.

[0110] Meanwhile, control information which is transmitted by a terminal to a base station through an uplink or is received by a terminal from a base station includes a downlink/uplink ACK/NACK (Acknowledgement/Non-Acknowledgement) signal, a CQI (Channel Quality Indicator), a PMI (Precoding Matrix Indicator), a RI (Rank Indicator), etc. For a 3GPP LTE system, a terminal may transmit control information of the above-described CQI/PMI/RI, etc. through a PUSCH and/or a PUCCH.

[0111] Table 5 represents an example of a DCI format in an NR system.

TABLE 5

DCI Format	Use
0_0	Scheduling of a PUSCH in one cell
0_1	Scheduling of one or multiple PUSCHs in one cell, or indication of cell group downlink feedback information to a UE
0_2	Scheduling of a PUSCH in one cell
1_0	Scheduling of a PDSCH in one DL cell
1_1	Scheduling of a PDSCH in one cell
1_2	Scheduling of a PDSCH in one cell

[0112] In reference to Table 5, DCI formats 0_0, 0_1 and 0_2 may include resource information (e.g., UL/SUL (Supplementary UL), frequency resource allocation, time resource allocation, frequency hopping, etc.), information related to a transport block (TB) (e.g., MCS (Modulation Coding and Scheme), a NDI (New Data Indicator), a RV (Redundancy Version), etc.), information related to a HARQ (Hybrid-Automatic Repeat and request) (e.g., a process number, a DAI (Downlink Assignment Index), PDSCH-HARQ feedback timing, etc.), information related to multiple antennas (e.g., DMRS sequence initialization information, an antenna port, a CSI request, etc.), power control information (e.g., PUSCH power control, etc.) related to scheduling of a PUSCH and control information included in each DCI format may be pre-defined.

[0113] DCI format 0_0 is used for scheduling of a PUSCH in one cell. Information included in DCI format 0_0 is CRC (cyclic redundancy check) scrambled by a C-RNTI (Cell Radio Network Temporary Identifier) or a CS-RNTI (Configured Scheduling RNTI) or a MCS-C-RNTI (Modulation Coding Scheme Cell RNTI) and transmitted.

[0114] DCI format 0_1 is used to indicate scheduling of one or more PUSCHs or configure grant (CG) downlink feedback information to a terminal in one cell. Information included in DCI format 0_1 is CRC scrambled by a C-RNTI or a CS-RNTI or a SP-CSI-RNTI (Semi-Persistent CSI RNTI) or a MCS-C-RNTI and transmitted.

[0115] DCI format 0_2 is used for scheduling of a PUSCH in one cell. Information included in DCI format 0_2 is CRC scrambled by a C-RNTI or a CS-RNTI or a SP-CSI-RNTI or a MCS-C-RNTI and transmitted.

[0116] Next, DCI formats 1_0, 1_1 and 1_2 may include resource information (e.g., frequency resource allocation, time resource allocation, VRB (virtual resource block)-PRB (physical resource block) mapping, etc.), information related to a transport block (TB) (e.g., MCS, NDI, RV, etc.), information related to a HARQ (e.g., a process number, DAI, PDSCH-HARQ feedback timing, etc.), information related to multiple antennas (e.g., an antenna port, a TCI (transmission configuration indicator), a SRS (sounding reference signal) request, etc.), information related to a PUCCH (e.g., PUCCH power control, a PUCCH resource indicator, etc.) related to scheduling of a PDSCH and control information included in each DCI format may be pre-defined.

[0117] DCI format 1_0 is used for scheduling of a PDSCH in one DL cell. Information included in DCI format 1_0 is CRC scrambled by a C-RNTI or a CS-RNTI or a MCS-C-RNTI and transmitted.

[0118] DCI format 1_1 is used for scheduling of a PDSCH in one cell. Information included in DCI format 1_1 is CRC scrambled by a C-RNTI or a CS-RNTI or a MCS-C-RNTI and transmitted.

[0119] DCI format 1_2 is used for scheduling of a PDSCH in one cell. Information included in DCI format 1_2 is CRC scrambled by a C-RNTI or a CS-RNTI or a MCS-C-RNTI and transmitted.

V2X (Vehicle-to-Everything)/Side Link (SL: Sidelink) Communication

[0120] FIG. 7 illustrates a procedure for performing V2X or SL communication according to a transmission mode in a wireless communication system to which the present disclosure may be applied.

[0121] The embodiment of FIG. 7 may be combined with various embodiments of the present disclosure. In various embodiments of the present disclosure, the transmission mode may be referred to as a mode or a resource allocation mode. Hereinafter, for convenience of explanation, the transmission mode in LTE may be referred to as an LTE transmission mode, and the transmission mode in NR may be referred to as an NR resource allocation mode.

[0122] For example, FIG. 7(a) illustrates the operation of a UE related to LTE transmission mode 1 or LTE transmission mode 3. Alternatively, for example, FIG. 7(a) illustrates the operation of a UE related to NR resource allocation mode 1. For example, LTE transmission mode 1 may be applied to general SL communication, and LTE transmission mode 3 may be applied to V2X communication.

[0123] For example, FIG. 7(b) illustrates the operation of a UE associated with LTE transmission mode 2 or LTE transmission mode 4. Alternatively, for example, FIG. 7(b) illustrates the operation of a UE associated with NR resource allocation mode 2.

[0124] Referring to FIG. 7(a), in LTE transmission mode 1, LTE transmission mode 3, or NR resource allocation mode 1, the base station may schedule SL resources to be used by the UE for SL transmission (S8000). For example, the base station may transmit information related to SL resources and/or information related to UL resources to the first UE. For example, the UL resources may include PUCCH resources and/or PUSCH resources. For example, the UL resources may be resources for reporting SL HARQ feedback to the base station.

[0125] For example, the first UE may receive information related to a dynamic grant (DG) resource and/or information related to a configured grant (CG) resource from the base station. For example, the CG resource may include a CG type 1 resource or a CG type 2 resource. In the present disclosure, the DG resource may be a resource that the base station configures/allocates to the first UE via DCI. In the present disclosure, the CG resource may be a (periodic) resource that the base station configures/allocates to the first UE via DCI and/or an RRC message. For example, in the case of a CG type 1 resource, the base station may transmit an RRC message including information related to the CG resource to the first UE. For example, in the case of a CG type 2 resource, the base station may transmit an RRC message including information related to the CG resource to the first UE, and the base station may transmit DCI related to activation or release of the CG resource to the first UE.

[0126] The first UE may transmit PSCCH (e.g., Sidelink Control Information (SCI) or 1st-stage SCI) to the second UE based on the resource scheduling (S8010).

[0127] The first UE may transmit a PSSCH (e.g., a 2nd-stage SCI, a MAC protocol data unit (PDU), data, etc.) related to the PSCCH to the second UE (S8020).

[0128] The first UE may receive a PSFCH related to the PSCCH/PSSCH from the second UE (S8030). For example, HARQ feedback information (e.g., NACK information or ACK information) can be received from the second UE via the PSFCH.

[0129] The first UE may transmit/report HARQ feedback information to the base station via PUCCH or PUSCH (S8040). For example, the HARQ feedback information reported to the base station may be information generated by the first UE based on HARQ feedback information received from the second UE. For example, the HARQ feedback information reported to the base station may be information generated by the first UE based on a rule set in advance. For example, the DCI may be DCI for scheduling of SL. For example, the format of the DCI may be DCI format 3_0 or DCI format 3_1.

[0130] Referring to FIG. 7(b), in LTE transmission mode 2, LTE transmission mode 4 or NR resource allocation mode 2, the UE may determine SL transmission resources within SL resources set by the base station/network or preset SL resources. For example, the set SL resources or preconfigured SL resources may be a resource pool. For example, the UE may autonomously select or schedule resources for SL transmission. For example, the UE may perform SL communication by selecting resources by itself within the configured resource pool. For example, the UE may perform sensing and resource (re)selection procedures to select resources by itself within a selection window. For example, the sensing may be performed on a sub-channel basis.

[0131] A first UE that has selected a resource within a resource pool may transmit a PSCCH (e.g., SCI or 1st-stage SCI) to a second UE using the resource (S8010).

[0132] The first UE may transmit a PSSCH (e.g., 2nd-stage SCI, MAC PDU, data, etc.) related to the PSCCH to the second UE (S8020).

[0133] The first UE may receive a PSFCH related to PSCCH/PSSCH from the second UE (S8030).

[0134] Referring to FIG. 7(a) or FIG. 7(b), for example, a first UE may transmit an SCI to a second UE on a PSCCH. Or, for example, the first UE may transmit two consecutive SCIs (e.g., 2-stage SCIs) to the second UE on the PSCCH and/or the PSSCH. In this case, the second UE may decode the two consecutive SCIs (e.g., 2-stage SCIs) to receive the PSSCH from the first UE. In the present disclosure, the SCI transmitted on the PSCCH may be referred to as a first (1st) SCI, a first SCI, a first-stage SCI, or a 1st-stage SCI format, and the SCI transmitted on the PSSCH may be referred to as a second (2nd) SCI, a second SCI, a second-stage SCI, or a 2nd-stage SCI format. For example, a 1st-stage SCI format may include SCI format 1-A, and a 2nd-stage SCI format may include SCI format 2-A and/or SCI format 2-B.

[0135] Referring to FIG. 7(a) or FIG. 7(b), in step S8030, the first UE may receive the PSFCH based on the description to be described below. For example, the first UE and the second UE may determine the PSFCH resource based on the description to be described below, and the second UE may transmit the HARQ feedback to the first UE using the PSFCH resource.

[0136] Describes the UE procedure for reporting HARQ-ACK on sidelink as specified in TS 38.213.

[0137] The UE may be indicated to transmit a PSFCH including HARQ-ACK information in response to the PSSCH reception by the SCI format scheduling the PSSCH

reception. The UE provides HARQ-ACK information including ACK or NACK or only NACK.

[0138] The UE may be provided with a number of slots within the resource pool during a period of PSFCH transmission occasion resources by sl-PSFCH-Period. If the number of slots is 0, the PSFCH transmission of the UE within the resource pool is disabled.

[0139] The UE expects that slot $t'_k{}^{SL}$ ($0 \leq k < T'_{max}$) includes PSFCH transmission occasion resources if $k \bmod N_{PSSCH}^{PSFCH} = 0$, where $t'_k{}^{SL}$ is defined in TS 38.214, T'_{max} is the number of slots belonging to the resource pool within 10240 msec according to TS 38.214, and N_{PSSCH}^{PSFCH} is provided by sl-PSFCH-Period.

[0140] The UE may be indicated by higher layers not to transmit a PSFCH containing HARQ-ACK information in response to receiving a PSSCH.

[0141] When a UE receives a PSSCH from a resource pool and the HARQ feedback enabled/disabled indicator field of the associated SCI format 2-A/2-B/2-C has value 1, the UE provides HARQ-ACK information in a PSFCH transmission from the resource pool. The UE transmits the PSFCH in the first slot that includes PSFCH resources and that corresponds to at least a number of slots provided by sl-MinTimeGapPSFCH of the resource pool after the last slot of the PSSCH reception.

[0142] The UE may be provided with a set of $M_{PRB, set}^{PSFCH}$ PRBs in the resource pool for PSFCH transmission including HARQ-ACK information in the physical resource block (PRB) of the resource pool by sl-PSFCH-RB-Set. The UE may be provided with a set of $M_{PRB, set}^{PSFCH}$ PRBs in the resource pool by sl-RB-SetPSFCH for PSFCH transmission including conflict information in the PRB of the resource pool. The UE expects that different PRBs are (pre-) set for conflict information and HARQ-ACK information. For the number of N_{SUBCH} sub-channels for the resource pool provided by sl-NumSubchannel and the number of PSSCH slots associated with PSFCH slots less than or equal to N_{PSSCH}^{PSFCH} , the UE allocates $[(i+j, N_{PSSCH}^{PSFCH}) M_{subch, slot}^{PSFCH} (i+1+j, N_{PSSCH}^{PSFCH}) M_{subch, slot}^{PSFCH} - 1]$ PRBs from $M_{PRB, set}^{PSFCH}$ PRBs among the PSFCH slots and PSSCH slots associated with sub-channel j to slot i . Here, $M_{subch, slot}^{PSFCH} = M_{PRB, set}^{PSFCH} / (N_{subch} N_{PSSCH}^{PSFCH})$, $0 \leq i < N_{PSSCH}^{PSFCH}$, $0 \leq j < N_{subch}$, and the allocation starts in ascending order of i and continues in ascending order of j . The UE expects $M_{PRB, set}^{PSFCH}$ to be a multiple of $N_{subch} N_{PSSCH}^{PSFCH}$.

[0143] The UE determines the number of available PSFCH resources for multiplexing HARQ-ACK or conflict information in PSFCH transmission based on the indication by sl-PSFCH-CandidateResourceType as $R_{PRB, CS}^{PSFCH} = N_{type}^{PSFCH} M_{subch, slot}^{PSFCH} N_{CS}^{PSFCH}$, where N_{CS}^{PSFCH} is the number of cyclic shift pairs for the resource pool provided by sl-NumMuxCS-Pair.

[0144] When sl-PSFCH-CandidateResourceType is set to startSubCH, $N_{type}^{PSFCH} = 1$, and $M_{subch, slot}^{PSFCH}$ PRBs are associated with the start sub-channel of the corresponding PSSCH.

[0145] When sl-PSFCH-CandidateResourceType is set to allocSubCH, $N_{type}^{PSFCH} = N_{subch}$, and $N_{PSSCH}^{PSFCH} M_{subch, slot}^{PSFCH}$ PRBs are associated with N_{subch} sub-channels of the corresponding PSSCH.

[0146] For conflict information, the corresponding PSSCH is determined based on sl-PSFCH-Occasion.

[0147] PSFCH resources are first indexed from $N_{type}^{PSFCH} M_{subch, slot}^{PSFCH}$ PRBs in ascending order of PRB index, and then from N_{CS}^{PSFCH} cyclic shift pairs in ascending order of cyclic shift pair index.

[0148] The UE determines the index of the PSFCH resource for transmitting the PSFCH including HARQ-ACK information or conflict information in response to the PSSCH reception, which corresponds to the reserved resource, as $(P_{ID} + M_{ID}) \bmod_{R_{PRB, CS}^{PSFCH}}$. Here, P_{ID} is a physical layer source identifier provided by SCI format 2-A/2-B/2-C that schedules PSSCH reception or SCI format 2-A/2-B/2-C that reserves resources for conflict information to be provided from other UEs. For HARQ-ACK information, if the UE detects SCI format 2-A having a cast type indicator field value of "01", M_{ID} is an identifier of the UE that receives the PSSCH indicated by the upper layer, otherwise M_{ID} is 0. For conflict information, M_{ID} is 0.

[0149] For PSFCH transmission including HARQ-ACK information or conflict information, the UE determines the value m_0 from the cyclic shift pair index corresponding to the PSFCH resource index and from N_{CS}^{PSFCH} using Table 6 below to calculate the value α of the cyclic shift.

[0150] Table 6 illustrates a set of cyclic shift pairs.

TABLE 6

N_{CS}^{PSFCH}	m_0					
	Cyclic shift pair index 0	Cyclic shift pair index 1	Cyclic shift pair index 2	Cyclic shift pair index 3	Cyclic shift pair index 4	Cyclic shift pair index 5
1	0	—	—	—	—	—
2	0	3	—	—	—	—
3	0	2	4	—	—	—
6	0	1	2	3	4	5

[0151] For PSFCH transmission including HARQ-ACK information, the UE determines a m_{cs} value as shown in Table 7 below if the UE detects an SCI format 2-A or SCI format 2-C having a cast type indicator field value of "01" or "10" to calculate the value α of the cyclic shift, and determines an m_{cs} value as shown in Table 8 below if the UE detects an SCI format 2-B or 2-A having a cast type indicator field value of "11". For PSFCH transmission including conflict information, the UE determines an m_{cs} value as shown in Table 9 below to calculate the value α of the cyclic shift. The UE applies one cyclic shift from a cyclic shift pair to a sequence used for PSFCH transmission. Table 7 illustrates the mapping of cyclic shifts of a sequence for PSFCH transmission from a cyclic shift pair and HARQ-ACK information bit values when the HARQ-ACK information includes ACK or NACK.

TABLE 7

HARQ-ACK value	0 (NACK)	1 (ACK)
Sequency cyclic shift	0	6

[0152] Table 8 illustrates the mapping of cyclic shifts of a sequence for PSFCH transmission from a cyclic shift pair and HARQ-ACK information bit values when HARQ-ACK information contains only NACK.

TABLE 8

HARQ-ACK value	0 (NACK)	1 (ACK)
Sequence cyclic shift	0	N/A

[0153] Table 9 illustrates the mapping of the cyclic shift of a sequence for PSFCH transmission and the value of the conflict information bit from a cyclic shift pair.

TABLE 9

Conflict information	Conflict information for the next time-reserved resource within SCI
Sequence cyclic shift	0

[0154] FIG. 8 illustrates a cast type for V2X or SL communication in a wireless communication system to which the present disclosure may be applied. The embodiment of FIG. 8 may be combined with various embodiments of the present disclosure. Referring to FIG. 8, FIG. 8(a) represents a broadcast type SL communication, FIG. 8(b) represents a unicast type SL communication, and FIG. 8(c) represents a group-cast type SL communication. In the case of a unicast type SL communication, a UE can perform one-to-one communication with another UE. In the case of a group-cast type SL communication, a UE may perform SL communication with one or more UEs within the group to which it belongs. In various embodiments of the present disclosure, SL group-cast communication may be referred to as SL multicast communication, SL one-to-many communication, etc.

[0155] Hereinafter, the HARQ (Hybrid Automatic Repeat Request) procedure is described.

[0156] For example, SL HARQ feedback may be enabled for unicast. In this case, in non-Code Block Group (CBG) operation, if a receiving UE decodes a PSCCH targeting the receiving UE and the receiving UE successfully decodes a transport block associated with the PSCCH, the receiving UE may generate an ACK. Then, the receiving UE may transmit the HARQ-ACK to the transmitting UE. On the other hand, if the receiving UE fails to successfully decode a transport block associated with the PSCCH after the receiving UE decodes the PSCCH targeting the receiving UE, the receiving UE may generate a NACK. Then, the receiving UE may transmit the NACK to the transmitting UE.

[0157] Also, for example, SL HARQ feedback may be enabled for group-cast. For example, in non-CBG operation, two HARQ feedback options may be supported for group-cast.

[0158] (1) group-cast option 1: If the receiving UE fails to decode a transport associated with the PSCCH after the receiving UE decodes the PSCCH targeting the receiving UE, the receiving UE may transmit a NACK to the transmitting UE via the PSFCH. On the other hand, if the receiving UE decodes the PSCCH targeting the receiving UE and the receiving UE successfully decodes the transport block associated with the PSCCH, the receiving UE may not transmit an ACK to the transmitting UE.

[0159] (2) Group-cast option 2: If the receiving UE fails to decode a transport block associated with the PSCCH after the receiving UE decodes the PSCCH targeting the receiving UE, the receiving UE may transmit a NACK to the trans-

mitting UE via the PSFCH. Then, if the receiving UE decodes the PSCCH targeting the receiving UE and the receiving UE successfully decodes the transport block associated with the PSCCH, the receiving UE may transmit an ACK to the transmitting UE via the PSFCH.

[0160] For example, if group-cast option 1 is used for SL HARQ feedback, all UEs performing group-cast communication may share PSFCH resources. For example, UEs belonging to the same group may transmit HARQ feedback using the same PSFCH resources.

[0161] For example, if group-cast option 2 is used for SL HARQ feedback, each UE performing group-cast communication may use different PSFCH resources for HARQ feedback transmission. For example, UEs belonging to the same group may transmit HARQ feedback using different PSFCH resources.

[0162] In the present disclosure, ACK may be referred to as HARQ-ACK, ACK information, or positive-ACK information, and NACK may be referred to as HARQ-NACK, NACK information, or negative-ACK information.

[0163] Sidelink grants may be semi-persistently configured by RRC or received dynamically on the PDCCH, which is autonomously selected by the MAC entity. The MAC entity may have a sidelink grant on the active SL BWP to determine the set of PSCCH periods in which SCI transmissions occur and the set of PSSCH periods in which SL-SCH transmissions associated with SCI occur.

[0164] When the MAC entity is configured to sidelink resource allocation mode 1, the MAC entity may perform actions based on each PDCCH occasion and each grant received for that PDCCH occasion.

UE/Base Station Behavior for Consistent LBT Failures in SL Transmission

[0165] NR Unlicensed Spectrum (NR-U) refers to a mode that provides the technology necessary for cellular operators to integrate unlicensed spectrum into their wireless communication systems. NR-U may enable both uplink and downlink operations in unlicensed spectrum.

[0166] In NR-U, channel access for uplink and downlink relies on LBT functionality. The UE and/or base station may first sense the communication channel to determine if there is no communication before transmission. If the communication channel is a wideband unlicensed carrier, the channel sensing procedure of NR-U may rely on sensing energy levels in multiple subbands of the communication channel. The base station may configure LBT parameters (e.g., type/duration, clear channel assessment parameters, etc.) for the UE.

[0167] In NR-U, a UE may determine the time and location of transmission and reception based on the indications of the COT (Channel Occupancy Time) structure. The COT consists of multiple slots, and each slot may consist of downlink resources, uplink resources, or flexible resources.

[0168] In an NR-U system, both contiguous and interlaced uplink resource allocations may be supported while complying with the regulations. In interlace uplink resource allocation, the basic unit of resource allocation for an NR unlicensed channel may be an interlace. For example, a single interlace may consist of ten equally spaced resource blocks within a 20 MHz frequency bandwidth for a 15 kHz subcarrier spacing.

[0169] For NR-U, the physical layer may perform the LBT procedure. The LBT procedure means a procedure in which

transmission is not performed when the channel is determined to be occupied. If the channel is identified as occupied according to the LBT procedure, transmission may not be performed by the physical layer. If the physical layer performs the LBT procedure before transmission and transmission is not performed, the physical layer may transmit an LBT failure indication to the MAC entity.

[0170] Unless otherwise specified, the operations specified in the present disclosure may be performed regardless of whether an LBT failure indication is received from the physical layer when an LBT procedure is performed for transmission. If the physical layer does not perform LBT, an LBT failure indication may not be received from the physical layer.

[0171] When SL transmission is supported in an unlicensed band, the UE may perform LBT for SL transmission. Here, LBT failure may occur not only in UL transmission but also in SL transmission. When SL mode 2 is configured, the UE may autonomously (re)select SL resources for SL transmission. In this case, when continuous LBT failures occur in SL transmission, a method for avoiding LBT failure for SL transmission (e.g., LBT failure for SL transmission when SL mode 2 is configured) is not defined in the basic wireless communication system.

[0172] Hereinafter, the operation of the terminal/base station when consecutive LBT failures occur for SL transmission is described.

[0173] In FIG. 9, for convenience of explanation, the sidelink Tx (transmission) UE is referred to as the first UE, and the sidelink Rx (reception) UE is referred to as the second UE. In addition, in FIG. 9, the sidelink operation between the first UE and the second UE may operate in an unlicensed band/spectrum or a shared spectrum.

[0174] FIG. 9 is a drawing for explaining a process in which a first UE performs communication according to one embodiment of the present disclosure.

[0175] The first UE may receive first configuration information related to a sidelink (SL)-specific LBT (listen before talk) failure detection timer (e.g., 'sl-LBT-FailureDetection-Timer') and second configuration information related to a SL-specific LBT failure counter from the base station (**S910**).

[0176] For example, the first UE may receive first configuration information and second configuration information included in upper layer signaling (e.g., RRC signaling) related to SL-specific LBT failure recovery from the base station.

[0177] For example, the second configuration information may include information related to a maximum number of SL-specific LBT failure instances for consistent SL-specific LBT detection (i.e., a SL-specific maximum LBT failure instance count threshold) (e.g., 'sl-LBT-FailureInstanceMaxCount') and/or a counter counting SL-specific failure indications (e.g., 'SL_LBT_COUNTER').

[0178] For example, a SL-specific failure detection timer and a SL-specific LBT failure counter may be configured for the first UE per serving cell.

[0179] Based on the SL-specific failure detection timer and the SL-specific LBT failure counter, the first UE may trigger a consistent SL-specific LBT failure (**S920**).

[0180] For example, a SL-specific LBT failure may be triggered based on a SL-specific failure detection timer and a SL-specific LBT failure counter for at least one of one or more RB sets or one or more resource pools.

[0181] Specifically, when an LBT failure indication is received from a lower layer, the SL-specific LBT failure counter value may be increased by 1 when the SL-specific failure detection timer is (re)started. If the SL-specific LBT failure counter value is greater than or equal to an SL-specific maximum LBT failure instance count threshold, the first terminal may trigger a consistent SL-specific LBT failure (for the activated SL BWP within the serving cell).

[0182] Here, based on at least one of the SL-specific failure detection timer and the SL-specific LBT failure counter being reset for the UE, the value of the SL-specific failure detection timer may be set to 0. Additionally, based on the expiration of the SL-specific failure detection timer, the value of the SL-specific failure detection timer may be set to 0.

[0183] The first UE may release the SL connection between the second UE and the first UE associated with a triggered consistent SL-specific LBT failure (**S930**).

[0184] For example, the SL connection between the second UEs may include a PC5-RRC connection and/or a unicast sidelink connection.

[0185] The first UE may transmit information to the base station indicating at least one of a triggered consistent SL-specific LBT failure or a release of the SL connection (**S940**).

[0186] For example, the first UE may transmit to the second UE a higher layer signaling (e.g., a PC5-RRC release message) including information for releasing the SL connection between the second UEs associated with the SL-specific LBT failure, thereby releasing the SL connection between the second UEs.

[0187] Additionally or alternatively, the first UE may transmit information indicating at least one of a triggered consistent SL-specific LBT failure or a release of an SL connection to the base station via PUCCH by triggering a scheduling request (SR).

[0188] As another example, the first UE may transmit information indicating release of the SL connection to the base station through upper layer signaling related to sidelink UE information (e.g., 'SidelinkUEInformation').

[0189] As another example, the first UE may transmit information indicating a triggered consistent SL-specific LBT failure to the base station via MAC CE. If the SL LBT failure granularity is SL BWP, the UE may declare an SL RLF (radio link failure) and the existing RRC message can be used to indicate SL RLF to all other UEs connected to it.

[0190] As another example, if the SL LBT failure granularity is a resource pool/RB set, then a UE may trigger SL RLF for all other terminals' connections if it triggers consistent SL LBT failures across all resource pools/RB sets.

[0191] The method performed by the first UE described in the example of FIG. 9 can be performed by the first device (**100**) of FIG. 13.

[0192] For example, one or more processors (**102**) of the first device (**100**) of FIG. 13 may be configured to receive first configuration information related to an SL-specific LBT failure detection timer and second configuration information related to an SL-specific LBT failure counter from a base station through one or more transceivers (**106**). The one or more processors (**102**) may be configured to trigger a consistent SL-specific LBT failure based on the SL-specific failure detection timer and the SL-specific LBT failure counter, and to release an SL connection between the second terminals related to the triggered consistent SL-specific LBT

failure. The one or more processors (102) may be configured to transmit information indicating at least one of the triggered consistent SL-specific LBT failure or the release of the SL connection to the base station through one or more transceivers (106).

[0193] Furthermore, one or more memories (104) of the first device (100) may store commands for performing the method described in the example of FIG. 9 when executed by one or more processors (102).

[0194] FIG. 10 is a drawing for explaining a process in which a base station performs communication according to one embodiment of the present disclosure.

[0195] The base station may transmit first configuration information related to the SL-specific LBT failure detection timer and second configuration information related to the SL-specific LBT failure counter to the first UE (S1010).

[0196] Features related to the first configuration information and the second configuration information have been described with reference to FIG. 9, so any duplicate description will be omitted.

[0197] The base station may receive information from the first UE indicating at least one of i) a consistent SL-specific LBT failure triggered by the first UE or ii) a release of an SL connection between the first UE and the second UE (S1020).

[0198] Here, a consistent SL-specific LBT failure may be triggered by the first UE based on the SL-specific failure detection timer and the SL-specific LBT failure counter. Then, the SL connection between the first UE and the second UE related to the triggered consistent SL-specific LBT failure may be released. That is, the base station may receive a report from the first UE that the SL connection between the first UE and the second UE is released due to the consistent SL-specific LBT failure.

[0199] The method performed by the base station described in the example of FIG. 10 can be performed by the second device (200) of FIG. 13.

[0200] For example, one or more processors (202) of the second device (200) of FIG. 13 may be configured to transmit first configuration information related to an SL-specific LBT failure detection timer and second configuration information related to an SL-specific LBT failure counter to the first terminal via one or more transceivers (206). The one or more processors (202) may be configured to receive, from the first UE via one or more transceivers (206), information indicating at least one of i) a consistent SL-specific LBT failure triggered by the first terminal or ii) a release of an SL connection between the first terminal and the second UE.

[0201] Furthermore, one or more memories (204) of the second device (200) may store instructions for performing the method described in the example of FIG. 10 when executed by one or more processors (202).

[0202] FIG. 11 is a diagram for explaining a method for performing resource reselection for a side link based on LBT failure according to one embodiment of the present disclosure.

[0203] The example of FIG. 11 is for convenience of explanation and does not limit the scope of the present disclosure. Some of the step(s) illustrated in FIG. 11 may be omitted depending on the situation and/or setting. In addition, the Tx terminal/Rx terminal/base station in FIG. 11 is only an example and may be implemented with the device illustrated in FIG. 13 below.

[0204] For example, the processor (102/202) of FIG. 13 may control to transmit and receive channels/signals/data/information, etc. using the transceiver (106/206). In addition, the processor (102/202) of FIG. 13 can control to store channels/signals/data/information, etc. to be transmitted, in the memory (104/204), and can also control to store received channels/signals/data/information, etc., in the memory (104/204).

[0205] The network (or base station) may transmit sidelink configuration information to a transmitting UE (i.e., Tx UE) via system information or/and terminal-specific RRC signaling (S1110).

[0206] When sidelink configuration information is received from the network, one or more carriers and one or more resource pools may be configured for each carrier for at least one of sidelink resource allocation mode 1 or mode 2 for the Tx UE.

[0207] Here, the number of HARQ processes may be configured based on each resource pool on each carrier. The resource pool may be used for uplink or sidelink transmission. Each resource pool may be associated with a pool index. The maximum number of HARQ processes for each resource pool may be configured. The number of HARQ retransmissions may correspond to the number of each HARQ process and/or the SL RSRP or DL RSRP measured by the (Tx) UE.

[0208] In the Tx UE, RRC may configure the MAC entity to sidelink resource allocation mode 2 to transmit using the resource pool(s) of the carrier based on detection or random selection. Then, the MAC entity (of the Tx UE) may perform each sidelink process.

[0209] For example, a TX UE may establish a PC5-RRC connection with a RX UE for unicast SL transmission.

[0210] At step S1120, the (Tx) UE checks transmission resource (re)selection, (re)selects transmission resources for generation of the selected sidelink grant, performs a channel access procedure with LBT type 1 or 2, and may generate a new MAC PDU for the selected sidelink grant within the interval of the COT.

[0211] If the MAC entity (of the Tx UE) has chosen to generate a selected sidelink grant corresponding to the transmission of multiple MAC PDUs and SL data is available on the logical channel: or

[0212] If the MAC entity chooses to generate a selected sidelink grant corresponding to transmission(s) of a single MAC PDU and SL data is available on the logical channel or SL-CSI reporting is triggered:

[0213] If the MAC entity does not select a resource pool allowed for the logical channel:

[0214] if 'sl-HARQ-FeedbackEnabled' is set to enabled for a logical channel, the (Tx) UE selects a resource pool consisting of PSFCH resources from among the resource pools;

[0215] if 'sl-HARQ-FeedbackEnabled' is not set to enabled for a logical channel, the (Tx) UE may select a specific resource pool from among the resource pools.

[0216] When the MAC entity in the (Tx) UE selects a new resource pool as described above or the MAC entity triggers Tx resource (re)selection, the MAC entity may perform TX resource (re)selection to generate the selected sidelink grant for the selected resource pool according to the method of transmission resource (re)selection described below.

[0217] The MAC entity may select one of the allowed values configured by RRC in 'sl-ResourceReservePe-

riodList' and set the resource reservation interval (P_{rsvp_TX}) to the selected value;

- [0218] The MAC entity may randomly select an integer value with equal probability from the interval [5, 15] if the resource reservation interval is greater than or equal to 100 ms, or from the interval

$$\left[5 \times \left\lceil \frac{100}{\max(20, P_{rsvp_TX})} \right\rceil, 15 \times \left\lceil \frac{100}{\max(20, P_{rsvp_TX})} \right\rceil \right]$$

if the resource reservation interval is less than 100 ms, and set 'SL_RESOURCE_RESELECTION_COUNTER' to the selected value;

- [0219] The MAC entity selects the number of HARQ retransmissions from the allowed number configured by RRC in 'sl-MaxTxTransNumPSSCH' included in 'sl-PSSCH-TxConfigList', and, if configured by RRC, it may be overlapped between 'sl-MaxTxTransNumPSSCH' indicated in 'sl-CBR-Priority TxConfigList' and the CBR measured by lower layers for the highest priority logical channel(s) allowed on the carrier if CBR measurement result is available, or the corresponding 'sl-defaultTxConfigIndex' configured by RRC if CBR measurement result is not available;
- [0220] The MAC entity selects an amount of frequency resources within the range configured by RRC among 'sl-MinSubChannelNumPSSCH' and 'sl-MaxSubchannelNumPSSCH' included in 'sl-PSSCH-TxConfigList', and if configured by RRC, it may be overlapped between 'MinSubChannelNumPSSCH' and 'MaxSubchannelNumPSSCH' indicated in 'sl-CBR-PriorityTxConfigList' and the CBR measured by lower layers, if there is a CBR measurement result, or the corresponding 'sl-defaultTxConfigIndex' configured by RRC.
- [0221] The MAC entity may arbitrarily select time and frequency resources for one transmission occasion from among the resources indicated by the physical layer, depending on the amount of frequency resources selected and the remaining PDB of SL data available on the logical channels allowed on the carrier;
- [0222] A MAC entity may use randomly selected resources to select a set of contiguous resources for transmission of PSCCH and PSSCH corresponding to multiple transmission occasions of multiple MAC PDUs or one or multiple transmission occasions of a single MAC PDU;
- [0223] If one or more HARQ retransmissions are selected and resources remain available for further transmission occasions as indicated by the physical layer;
- [0224] The MAC entity may randomly select time and frequency resources for one or more transmission occasions from the available resources, depending on the remaining PDB of SL data available on logical channels allowed in the carrier, ensuring the amount of selected frequency resources, the number of selected HARQ retransmissions and the minimum time interval between two selected resources. The above operation may be performed when the PSFCH is set for the corresponding resource pool and the retransmission resources may be indicated by the time resource allocation of the previous SCI:

- [0225] A MAC entity may select a set of periodic resources spaced by a resource reservation interval for PSCCH and PSSCH transmissions corresponding to the number of retransmission opportunities of the determined MAC PDU using randomly selected resources;

- [0226] A MAC entity may consider the first set of transmission occasions as initial transmission occasions and may consider other sets(s) of transmission occasions as retransmission occasions;

- [0227] The MAC entity may consider the set of initial transmission occasions and retransmission occasions as the selected sidelink grants.

- [0228] If one or more HARQ retransmissions are not selected, the MAC entity may consider the set as the selected sidelink grant.

- [0229] The MAC entity may use the selected sidelink grant to determine the PSCCH duration set and the PSSCH duration set.

- [0230] Alternatively, at the Tx UE, when the MAC entity selects a new resource pool or the MAC entity triggers TX resource (re)selection, the MAC entity may perform TX resource (re)selection to generate the selected sidelink grant for the selected resource pool according to the following manner of Tx resource (re)selection:

- [0231] The MAC entity selects the number of HARQ retransmissions from the allowed number configured by RRC in 'sl-MaxTxTransNumPSSCH' included in 'sl-PSSCH-TxConfigList', and if configured by RRC, it may be overlapped between 'sl-MaxTxTransNumPSSCH' indicated in 'sl-CBR-Priority TxConfigList' and the CBR measured in lower layers, if CBR measurement result is not available, and the corresponding 'sl-defaultTxConfigIndex' configured by RRC.

- [0232] The MAC entity selects an amount of frequency resources within the ranges configured by RRC among 'sl-MinSubChannelNumPSSCH' and 'sl-MaxSubchannelNumPSSCH' included in 'sl-PSSCH-TxConfigList', and if set by RRC, it may be overlapped between 'MinSubChannelNumPSSCH' and 'MaxSubchannelNumPSSCH' indicated in 'sl-CBR-Priority TxConfigList' and the CBR measured in lower layers, if CBR measurement result is not available, and the corresponding 'sl-defaultTxConfigIndex' set by RRC;

- [0233] The MAC entity may arbitrarily select time and frequency resources for one transmission occasion from among the resources indicated by the physical layer, depending on the amount of frequency resources selected and the remaining PDB of SL data available on the logical channels allowed on the carrier.

- [0234] A MAC entity may use randomly selected resources to select a set of contiguous resources for transmission of PSCCH and PSSCH corresponding to multiple transmission occasions of multiple MAC PDUs or one or multiple transmission occasions of a single MAC PDU.

- [0235] For transmission of PSCCH and PSSCH corresponding to one of the multiple transmission occasions of multiple MAC PDUs, the MAC entity may select a series of consecutive resources periodically based on a resource reservation interval.

- [0236] Here, one of the allowed values configured by RRC in 'sl-ResourceReservePeriodList' may be selected as the

resource reservation interval. For each resource reservation interval, a set of contiguous resources may be reserved.

[0237] One or more HARQ processes may be allocated to a single contiguous resource set in each resource reservation interval. The number of HARQ processes may be determined by the (maximum or fixed) number of HARQ processes for each contiguous resource set or for each resource reservation interval.

[0238] The number of (maximum or fixed) HARQ processes for each contiguous resource set or each resource reservation interval may be configured by the base station or preconfigured/defined.

[0239] The (maximum or fixed) number of HARQ processes may be (pre-) configured per resource pool, per SL BWP, per serving cell, per selected sidelink grant or per logical channel priority of the SL logical channel used to trigger resource selection.

[0240] For each consecutive resource set or resource reservation interval, for one or more HARQ process IDs indicated in one or more SCIs, different SCIs may indicate the same or different HARQ process IDs, as described below, to schedule different PSSCH transmissions in the same TB or in different TBs.

[0241] For a plurality of TBs transmitted to the same destination UE, the UE may assign different HARQ process IDs to different HARQ processes for each consecutive resource set or resource reservation interval.

[0242] When a plurality of TBs are transmitted to different destination UEs, the UE may allocate the same or different HARQ process IDs to different HARQ processes for each consecutive resource set or resource reservation interval.

[0243] One or more PSFCH resources may be reserved for each contiguous resource set or resource reservation interval.

[0244] Option 1: One or more PSFCH resources in a resource reservation interval may be used to send one or more HARQ-ACKs on a series of consecutive PSSCH resources that are always in the same resource reservation interval.

[0245] Option 2: One or more PSFCH resources in a resource reservation interval may be used to transmit one or more HARQ-ACKs on a series of consecutive PSSCH resources in a previous resource reservation interval.

[0246] Other PSFCH resources in the resource reservation interval may be used to transmit one or more HARQ-ACKs to consecutive PSSCH resource sets in the same resource reservation interval.

[0247] The MAC entity may consider the set as the selected sidelink grant. The MAC entity may use the selected sidelink grant to determine the PSSCH period set and the PSSCH period set.

[0248] Thereafter, in the steps described later, the UE may perform SL transmission on the selected sidelink grant using the PSSCH duration set and the PSSCH duration set.

[0249] In S1130, the Tx UE may perform sidelink (re) transmission of MAC PDU on the selected resource and receive SL ACK/NACK on the PSFCH. Here, if SL NACK is received on the PSFCH, the Tx UE may (re)select the Tx resource or reserve additional resources for retransmission of MAC PDU.

[0250] A consistent (or continuous) LBT failure recovery procedure may be established by RRC for the MAC entity of the (Tx) UE. If the physical layer of the UE fails to access

the channel(s) prior to the intended UL or SL transmission to the base station, the physical layer may notify the MAC layer of the channel access failure (i.e., LBT failure).

[0251] The UE may detect/detect consistent (or continuous) LBT failure per SL BWP or per UL BWP or per serving cell by counting LBT failure indications for all SL transmissions and/or all UL transmissions from the physical layer to the MAC entity based on at least one of the embodiments described below (S1140).

[0252] The Tx UE may transmit a PUCCH for SR to the base station (S1150). At step S1160, the Tx UE may transmit a message/information for SL connection release to the Rx terminal, and the Rx UE may transmit an SL ACK to the Tx UE on the PSFCH. For example, based on the detection of consistent SL LBT failure, the Tx UE may transmit a message for SL connection release to the Rx UE.

[0253] The Tx UE may transmit sidelink UE information (including information related to disconnection based on LBT failure) to the base station (S1170).

[0254] Hereinafter, the above-described operation of the Tx UE is described in detail.

Embodiment 1: Integrated UL/SL LBT Failure Detection and Recovery Procedure

[0255] The UE may count LBT_COUNTER based on LBT failure for both UL transmission via UL BWP and SL transmission via SL BWP for each serving cell. LBT failure MAC CE may be generated and shared for LBT failure in both UL and SL. SR (scheduling request) may be triggered and shared for LBT failure in both UL and SL. SR configuration and resource(s) may be common for both UL LBT failure reporting and SL LBT failure reporting. Parameters of 'lbt-FailureRecoveryConfig' (e.g., 'lbt-FailureInstanceMaxCount' and 'lbt-FailureDetectionTimer') may be shared between UL and SL, and may be configured by RRC.

[0256] A consistent LBT failure recovery procedure for the MAC entity of the UE may be established by RRC. Consistent LBT failure per UL BWP, per SL BWP, or per BWP overlapping a UL BWP and an SL BWP may be detected by counting LBT failure indications for all UL transmissions and/or all SL transmissions from the lower layer (i.e., the physical layer) to the MAC entity.

[0257] LBT_COUNTER (per serving cell) may be used for consistent LBT failure detection procedure. The LBT_COUNTER is a counter for LBT failure indications that is initially set to 0. For each serving cell, LBT_COUNTER may be used to count LBT failure indications for UL transmissions only for the UL BWP of the serving cell, or LBT failure indications for SL transmissions only for the SL BWP of the serving cell, or both LBT failure indications for UL transmissions and LBT failure indications for SL transmissions for both the UL BWP and SL BWP of the serving cell.

[0258] For each activated serving cell configured with 'lbt-FailureRecoveryConfig', the MAC entity of the UE may perform the following actions:

[0259] 1> if LBT failure indication has been received from lower layers:

[0260] 2> start or restart lbt-FailureDetection Timer;

[0261] 2> increment LBT_COUNTER by 1;

[0262] 2> if LBT_COUNTER ≥ lbt-FailureInstanceMaxCount:

[0263] 3> trigger consistent LBT failure for active UL BWP on the serving cell;

- [0264] 3> trigger consistent LBT failure for active SL BWP on the serving cell;
- [0265] 3> if the serving cell is a SpCell:
- [0266] 4> if consistent LBT failure is triggered on all UL BWPs set to PRACH opportunities on the same carrier of the serving cell; or
- [0267] 4> if a consistent LBT failure is triggered on the corresponding SL BWP and a PRACH occasion is configured on the UL BWP of the same carrier as the SL BWP in the corresponding serving cell:
- [0268] 5> indicate the consistent LBT failure to the higher layer (i.e., RRC layer).
- [0269] 4> else:
- [0270] 5> stop the ongoing random access procedure on the serving cell;
- [0271] 5> switch the active UL BWP to the (target) UL BWP on the same carrier of the serving cell that is configured with a PRACH occasion and for which no consistent LBT failure has been triggered;
- [0272] 5> switch the active SL BWP to the (target) SL BWP on the same carrier of the serving cell that is configured with a PRACH occasion and for which no consistent LBT failure has been triggered, wherein the target SL BWP is associated with or overlaps the target UL BWP.
- [0273] 5> initiate the random access procedure.
- [0274] 1> if all triggered consistent LBT failures are canceled on the serving cell: or
- [0275] 1> if lbt-FailureDetectionTimer expires; or
- [0276] 1> If lbt-FailureDetectionTimer or lbt-Failure-InstanceMaxCount is reconfigured by a higher layer:
- [0277] 2> set LBT_COUNTER to 0.
- [0278] The MAC entity of the UE shall:
- [0279] 1> if a consistent LBT failure is triggered and not canceled in SpCell; and
- [0280] 1> if UL-SCH resources are available for a new transmission in SpCell and the UL-SCH resources may accommodate an LBT failure MAC CE and its sub-headers as a result of logical channel priorities:
- [0281] 2> indicate the multiplexing and assembly procedure to generate an LBT failure MAC CE.
- [0282] 1> if a consistent LBT failure is triggered and not canceled on at least one SCell:
- [0283] 2> if UL-SCH resources are available for new transmission in a serving cell where consistent LBT failure has not been triggered and the UL-SCH resources can accommodate the LBT failure MAC CE and its subheaders as a result of logical channel priority:
- [0284] 3> indicate the multiplexing and assembly procedure to generate an LBT failure MAC CE.
- [0285] 2> else:
- [0286] 3> trigger scheduling request for LBT failure MAC CE.
- [0287] 1> when a MAC PDU is transmitted and no LBT failure indication is received from a lower layer and the PDU contains an LBT failure MAC CE:
- [0288] 2> cancel all consistent LBT failure(s) triggered on SCell(s) indicated by the transmitted LBT failure MAC CE.
- [0289] 1> if a consistent LBT failure is triggered and not cancelled in SpCell; and
- [0290] 1> if a random access procedure is considered to have completed successfully in SpCell:

- [0291] 2> cancel all consistent LBT errors triggered in SpCell.
- [0292] 1> if lbt-FailureRecoveryConfig is reconfigured by the upper layer for the serving cell:
- [0293] 2> cancel all consistent LBT failures triggered on that serving cell.

Embodiment 1-1

[0294] Embodiment 1-1 relates to LBT failure MAC CE for Example 1.

[0295] A one-octet LBT failure MAC CE may be identified by a MAC subheader having a specific LCID (e.g., an LCID mapped to index/codepoint 43 or/and 49). As illustrated in (a) of FIG. 12, the size of a one-octet LBT failure MAC CE is fixed and may consist of a single octet including eight C fields.

[0296] A 4-octet LBT failure MAC CE may be identified by a MAC subheader having a specific LCID (e.g., an LCID mapped to index/codepoint 42 or/and 48). As illustrated in (b) of FIG. 12, the size of a 4-octet LBT failure MAC CE is fixed and may consist of 4 octets containing 32 C fields.

[0297] If the highest 'ServCellIndex' of the serving cell of the MAC entity where the LBT failure was detected is less than 8, the single octet format is used, otherwise the four octet format may be used.

[0298] If there is a serving cell configured for a MAC entity with 'ServCellIndex' i in (a) and (b) of FIG. 12, and a consistent LBT failure is triggered across UL and SL and the consistent LBT failure is not canceled on that serving cell, Ci may be set to 1. Otherwise, Ci may be set to 0.

Embodiment 2: SL Specific LBT Failure Detection and Recovery Procedure

[0299] That is, Embodiment 2 relates to a separate LBT failure detection and recovery procedures for SL LBT failure and UL LBT failure, and possibly the interaction between SL LBT failure and UL LBT failure.

[0300] The UE may separately count the existing LBT_COUNTER based on LBT failures for UL transmissions in the UL BWP and the new LBT_COUNTER_SL based on LBT failures for SL transmissions in the SL BWP for each serving cell.

[0301] The existing LBT failure MAC CE and the new LBT failure MAC CE may be generated separately for LBT failure of UL and LBT failure of SL, respectively. SR for UL LBT failure and SR for SL LBT failure may be triggered separately. SR configuration and resources may be separately configured and allocated for UL LBT failure and SL LBT failure reporting.

[0302] Parameters of 'lbt-FailureRecoveryConfig' (e.g., 'lbt-FailureInstanceMaxCount' and 'lbt-FailureDetectionTimer') may be individually configured via RRC for UL LBT failure and SL LBT failure respectively.

[0303] Additionally or alternatively, as embodiment 2, the existing LBT failure MAC CE may be reused for SL LBT fault reporting and UL LBT failure reporting. That is, the same MAC CE format may be used to indicate SL LBT failure or UL LBT failure based on different LC (logical channel) ID values.

[0304] Parameters in 'lbt-FailureRecoveryConfig' (e.g., 'lbt-FailureInstanceMaxCount' and 'lbt-FailureDetectionTimer') may be reused for SL LBT failure detection and UL LBT failure detection. That is, 'lbt-FailureInstanceMax-

Count' or 'lbt-FailureDetectionTimer' may operate based on the configuration of the same corresponding parameter in 'lbt-FailureRecoveryConfig'.

[0305] Here, the SR configurations and resources may be common to both UL LBT failure reporting and SL LBT failure reporting.

[0306] Since the BWP may be composed of one or more RB sets (i.e., 20 MHz LBT bandwidth) on which the UE performs LBT, the UE may perform LBT failure detection and recovery procedures for SL or UL specific RB sets instead of SL BWP or UL BWP. In this case, the BWP may be replaced with the RB set in each embodiment.

[0307] The MAC entity of the UE may be configured by RRC with a consistent LBT failure recovery procedure. Consistent LBT failures per UL BWP, per SL BWP, or per BWP overlapping a UL BWP and an SL BWP may be detected by counting LBT failure indications for all UL transmissions or all SL transmissions from lower layers to the MAC entity.

[0308] RRC may configure the following parameters in 'lbt-FailureRecoveryConfig' for UL LBT failure detection.

[0309] lbt-FailureInstanceMaxCount for consistent LBT failure detection in UL

[0310] lbt-FailureDetectionTimer for consistent LBT failure detection in UL

[0311] RRC may configure the following parameters in 'lbt-FailureRecoveryConfigSL' to detect SL LBT failure.

[0312] 'lbt-FailureInstanceMaxCountSL' for consistent LBT failure detection in SL

[0313] 'lbt-FailureDetectionTimerSL' for consistent LBT error detection in SL

[0314] The following UE parameters may be used for a consistent LBT failure detection procedure for UL LBT failure.

[0315] LBT_COUNTER (per serving cell): Counter for LBT failure indications, initially set to 0 for UL.

[0316] The following UE parameters may be used for a consistent LBT failure detection procedure for SL LBT failures.

[0317] LBT_COUNTER_SL (per serving cell): Counter for LBT failure indication, initially set to 0 for SL.

[0318] For each activated serving cell set to 'lbt-FailureRecoveryConfig', the MAC entity of the UE may detect a UL LBT failure:

[0319] 1> if an LBT failure indication is received from a lower layer for UL transmission:

[0320] 2> start or restart lbt-FailureDetectionTimer;

[0321] 2> increment LBT_COUNTER by 1;

[0322] 2> if LBT_COUNTER >= lbt-FailureInstanceMaxCount:

[0323] 3> trigger consistent LBT failure for active UL BWP on the serving cell;

[0324] 3> if the serving cell is a SpCell:

[0325] 4> if consistent LBT failure is triggered in all UL BWPs configured as PRACH occasions on the same carrier of the serving cell:

[0326] 5> indicate consistent LBT failure to the upper layer (i.e. RRC layer).

[0327] 4> else:

[0328] 5> stop the ongoing random access procedure on the serving cell;

[0329] 5> switch the active UL BWP to the (target) UL BWP on the same carrier of the serving cell that is

configured with a PRACH occasion and for which no consistent LBT failure is triggered;

[0330] 5> switch the active SL BWP to the (target) SL BWP on the same carrier of the serving cell that is not triggered to a consistent LBT failure, wherein the target SL BWP is associated with or overlaps the target UL BWP;

[0331] 5> initiate the random access procedure.

[0332] 1> if LBT_COUNTER_SL is set to 0 for an SL LBT failure; or

[0333] 1> if all consistent LBT failures triggered in an SL transfer are canceled on the serving cell; or

[0334] 1> if lbt-FailureDetectionTimerSL expires; or

[0335] 1> if lbt-FailureDetectionTimerSL or lbt-FailureInstanceMaxCountSL is reconfigured by an upper layer; or

[0336] 1> if all consistent LBT failures triggered are canceled on the serving cell; or

[0337] 1> if lbt-FailureDetectionTimer expires; or

[0338] 1> if lbt-FailureDetectionTimer or lbt-FailureInstanceMaxCount is reconfigured by an upper layer:

[0339] 2> Set LBT_COUNTER to 0.

[0340] for each activated serving cell configured with lbt-FailureRecoveryConfigSL, the MAC entity of the UE may detect SL LBT failures.

[0341] 1> if an LBT failure indication is received from a lower layer for SL transmission:

[0342] 2> start or restart lbt-FailureDetectionTimerSL;

[0343] 2> increment LBT_COUNTER_SL by 1;

[0344] 2> if LBT_COUNTER_SL >= lbt-FailureInstanceMaxCountSL:

[0345] 3> trigger consistent LBT failure for active SL BWP on the serving cell;

[0346] 3> if the serving cell is a SpCell:

[0347] 4> if consistent LBT failure is triggered on all SL BWPs and PRACH triggering is configured on UL BWP of the same carrier as the SL BWP in that serving cell:

[0348] 5> indicates a consistent LBT failure to the upper layer (i.e., RRC layer).

[0349] 4> else;

[0350] 5> stop any random access procedure in progress in the serving cell;

[0351] 5> switch the active SL BWP to the (target) SL BWP on the same carrier of the serving cell where the UL BWP is configured with a PRACH occasion and no consistent LBT failure is triggered in UL or SL.

[0352] 5> switch the active UL BWP to the (target) UL BWP on the same carrier of the serving cell where no consistent LBT failure is triggered, wherein the target UL BWP is associated with or overlaps the target SL BWP;

[0353] 5> initiating a random access procedure.

[0354] 1> if LBT_COUNTER is set to 0 for a UL LBT failure; or

[0355] 1> if lbt-FailureDetectionTimer expires for a UL LBT failure; or

[0356] 1> if lbt-FailureDetectionTimer or lbt-FailureInstanceMaxCount is reconfigured by upper layers on UL LBT failure:

[0357] 1> if all consistent LBT failures triggered in a UL transmission are canceled on the serving cell; or

[0358] 1> if all consistent LBT failures triggered in a SL transmission are canceled on the serving cell; or

[0359] 1> if lbt-FailureDetectionTimerSL expires; or

[0360] 1> if lbt-FailureDetectionTimerSL or lbt-FailureInstanceMaxCountSL is reconfigured by an upper layer;

[0361] 2> Set LBT_COUNTER_SL to 0.

[0362] For reporting a UL LBT failure, the MAC entity of the terminal may perform the following actions:

[0363] 1> If a consistent LBT failure in the SpCell is triggered and not cancelled for UL; and

[0364] 1> If UL-SCH resources are available for new transmission in the SpCell and these UL-SCH resources may accommodate the LBT failure MAC CE and its subheaders as a result of logical channel prioritization;

[0365] 2> indicate the multiplexing and assembly procedure to generate an LBT failure MAC CE.

[0366] 1> if a consistent LBT failure on at least one SCell is triggered for UL and not cancelled;

[0367] 2> If UL-SCH resources are available for a new transmission in a serving cell where consistent LBT failures for UL or SL have not been triggered, and if those UL-SCH resources may accommodate the LBT failure MAC CE and its subheaders as a result of logical channel prioritization;

[0368] 3> indicate the multiplexing and assembly procedure to generate a LBT failure MAC CE.

[0369] 2> else;

[0370] 3> trigger scheduling request for LBT failure MAC CE.

[0371] 1> when a MAC PDU is transmitted and no LBT failure indication is received from a lower layer and the PDU includes an LBT failure MAC CE;

[0372] 2> cancel all consistent LBT failure(s) triggered on SCell(s) that indicated consistent LBT failure in transmitted LBT failure MAC CE.

[0373] 1> if a consistent LBT failure is triggered for UL and is not canceled in SpCell; and

[0374] 1> if a random access procedure is considered to have completed successfully in SpCell;

[0375] 2> (Alt 1-1) cancel all consistent LBT failures triggered on UL except SL in SpCell.

[0376] 2> (Alt 1-2) cancel all consistent LBT failures triggered on both UL and SL in SpCell.

[0377] 1> if lbt-FailureRecoveryConfig is reconfigured by upper layers for the serving cell;

[0378] 2> cancel all consistent LBT failures triggered for ULs on that serving cell.

[0379] 2> Cancel all consistent LBT failures triggered for SLs on that serving cell.

[0380] For reporting SL LBT failure, the MAC entity of the terminal may perform the following actions:

[0381] 1> if a consistent LBT failure in the SpCell is triggered for the SL and has not been cancelled; and

[0382] 1> if UL-SCH resources are available for new transmission in the SpCell and the UL-SCH resources may accommodate the SL LBT failure MAC CE and its subheader as a result of logical channel prioritization;

[0383] 2> indicate the multiplexing and assembly procedure to generate a SL LBT failure MAC CE

[0384] 1> if a consistent LBT failure has been triggered for the SL on at least one SCell and has not been cancelled;

[0385] 2> if UL-SCH resources are available for new transmission in the serving cell where a consistent LBT

failure has not been triggered for the UL or SL, and the UL-SCH resources may accommodate the SL LBT failure MAC CE and its subheaders as a result of logical channel prioritization;

[0386] 3> indicate the multiplexing and assembly procedure to generate a SL LBT failure MAC CE.

[0387] 2> else;

[0388] 3> trigger scheduling request for SL LBT failure MAC CE.

[0389] 1> If a MAC PDU is transmitted and no SL LBT failure indication is received from the lower layer and the PDU includes a SL LBT failure MAC CE;

[0390] 2> Cancel all consistent LBT failure(s) triggered on SCell(s) that indicated consistent LBT failure in transmitted LBT failure MAC CE.

[0391] 1> if a consistent LBT failure is triggered for SL and is not canceled in SpCell; and

[0392] 1> if the Random Access procedure is considered to have completed successfully in SpCell;

[0393] 2> (Alt 2-1) cancel all consistent LBT failures triggered for SL except UL in SpCell.

[0394] 2> (Alt 2-2) cancel all consistent LBT failures triggered for both UL and SL in SpCell.

[0395] 1> if lbt-FailureRecoveryConfigSL is reconfigured by the upper layer for the serving cell;

[0396] 2> cancel all consistent LBT failures triggered for SLs on that serving cell

[0397] 2> cancel all consistent LBT failures triggered for ULs on that serving cell

Embodiment 2-1

[0398] Embodiment 2-1 relates to LBT failure MAC CE and SL LBT failure MAC CE for Embodiment 1.

[0399] One octet LBT failure MAC CE and SL LBT failure MAC CE may be identified by a MAC subheader with a specific LCID. As illustrated in (a) of FIG. 12, the size of one octet LBT failure MAC CE is fixed and may be composed of a single octet including eight C fields.

[0400] 4-octet LBT failure MAC CE and SL LBT failure MAC CE may be identified by a MAC subheader with a specific LCID. As shown in (b) of FIG. 12, the size of the 4-octet LBT failure MAC CE is fixed and can be composed of 4 octets including 32 C fields.

[0401] The UCID value for UL-SCH may be defined as shown in Table 10 below.

TABLE 10

Codepoint/Index	LCID value
42	SL LBT failure (4 Octet)
43	SL LBT failure (1 Octet)
48	UL LBT failure (4 Octet)
49	UL LBT failure (1 Octet)

[0402] If the highest 'ServCellIndex' of the serving cell of the MAC entity where the LBT failure is detected is less than 8, the single octet format is used, otherwise the four octet format may be used. In (a) and (b) of FIG. 12, if there is a serving cell configured for the MAC entity with 'ServCellIndex' i and a consistent LBT failure is triggered across UL and SL and the consistent LBT failure is not canceled in that serving cell, Ci may be set to 1. Otherwise, Ci may be set to 0.

[0403] Whether embodiment 1 or embodiment 2 is used may depend on the SL mode or the serving cell where SL transmission is performed. For example, embodiment 1 may be used in SL mode 1, and embodiment 2 may be used in SL mode 2. Embodiment 1 may be used in a serving cell where both UL and SL transmissions are allowed, and embodiment 2 may be used in a serving cell. In this case, only SL transmission may be allowed, and UL transmission may not be allowed.

Embodiment 3

[0404] Embodiment 3 relates to the operation of a Tx terminal performed subsequent to Embodiment 1 and/or Embodiment 2.

[0405] At a TX UE, if a selected sidelink grant is available for retransmission(s) of a positively acknowledged MAC PDU, the UE may clear the PSCCH period(s) and PSSCH period(s) corresponding to the retransmission(s) of the MAC PDU in the selected sidelink grant. Here, the selected sidelink grant may correspond to transmission of a single MAC PDU or transmission of multiple MAC PDUs.

[0406] At the TX UE, the MAC entity may continuously perform the TX resource (re)selection confirmation procedure until the corresponding resource pool is released by RRC or the MAC entity decides to cancel the generation of the selected sidelink grant corresponding to the transmission of multiple MAC PDUs or the transmission of a single MAC PDU.

[0407] If one or more of the following conditions are met, the MAC entity may perform a TX resource (re)selection check from the selected resource pool for the corresponding sidelink process. The condition(s) are as follows:

[0408] If SL_RESOURCE_RESELECTION_COUNTER=0 and SL_RESOURCE_RESELECTION_COUNTER is 1, the MAC entity may randomly select a value in the interval [0, 1] with an equal probability higher than the probability configured by RRC in 'sl-ProbResourceKeep', if the following options apply:

[0409] Option 1: If the transmission corresponds to the last transmission of a MAC PDU, the UE may decrement the SL_RESOURCE_RESELECTION_COUNTER by 1 (if available) even if the last transmission was not transmitted due to an SL LBT failure.

[0410] Option 2: If a transmission of a MAC PDU is not performed due to an LBT failure when the terminal determines the last transmission of a MAC PDU for the SL_RESOURCE_RESELECTION_COUNTER, the failed transmission may not be included in the number of transmissions of the MAC PDU.

[0411] 'sl-ProbResourceKeep' may be scaled up or down based on consistent LBT failures for the SL. A certain number of LBT failures in a selected sidelink grant or SL BWP may increase or decrease the value of 'sl-ProbResourceKeep'. For example, by multiplying 'sl-ProbResourceKeep' by N/10, N may be determined by the number of LBT failures or triggered consistent LBT failures.

[0412] if the resource pool is established or reconfigured by RRC: or

[0413] if there is no selected sidelink grant in the selected resource pool: or

[0414] if the resources of the selected sidelink grant are not within the COT (Channel Occupancy Time) duration determined by the channel access procedure performed by the UE or shared by any UE/gNB/eNB: or

[0415] if a consistent LBT failure is triggered for the selected sidelink grant: or

[0416] if a consistent LBT failure is triggered for the active SL BWP of the corresponding serving cell associated with the sidelink grant selected in the above step: or

[0417] if consistent LBT failure is triggered in all SL BWPs of the serving cell associated with the selected sidelink grant: or

[0418] if a consistent LBT failure is triggered and not cancelled for SL or UL, in the corresponding serving cell associated with the selected sidelink grant: or

[0419] if no transmission or retransmission is actually performed by the MAC entity on the resources indicated in the selected sidelink grant during the last 1 second and the following alternatives apply; or

[0420] Alternative 1: if a MAC PDU transmission is not performed due to an LBT failure, the failed transmission may not be considered a real (re)transmission.

[0421] Alternative 2: if a MAC PDU transmission is not performed due to an LBT failure, the failed transmission may be considered as an actual (re)transmission.

[0422] if 'sl-ReselectAfter' is configured and the number of consecutive unused transmission occasions for the resource indicated in the selected sidelink grant is equal to 'sl-ReselectAfter' and the following alternatives apply: or

[0423] Alternative 1: If a MAC PDU transmission is not performed due to an LBT failure, the transmission occasion of the failed transmission may not be considered as an unused transmission occasion for 'sl-ReselectAfter'.

[0424] Alternative 2: If a MAC PDU transmission is not performed due to an LBT failure, the transmission occasion of the failed transmission may be considered as an unused transmission occasion for 'sl-ReselectAfter'.

[0425] if the selected sidelink grant cannot accommodate the RLC SDU using the maximum allowable MCS configured by RRC in 'sl-MaxMCS-PSSCH' and the UE chooses not to segment the RLC SDU; or

[0426] if the transmission(s) with the selected sidelink grant cannot satisfy the latency requirements of the data on the logical channel according to the associated priority and the MAC entity chooses not to perform the transmission(s) corresponding to a single MAC PDU:

[0427] if the TX UE receives an SL NACK in response to a transmission of a MAC PDU at the last transmission occasion within the COT duration determined by the channel access procedure performed by the UE or shared by any UE/gNB/eNB for a sidelink grant selected by the UE, and one or more retransmissions of the MAC PDU are still required (e.g., due to the maximum number of transmissions of the MAC PDU); or

[0428] if one or more retransmissions of a MAC PDU are still required after the last transmission occasion within the COT duration determined by the channel access procedure performed by the UE or shared by any UE/gNB/eNB for the selected sidelink grant (e.g. due to the maximum number of transmissions of a MAC PDU with HARQ feedback disabled),

[0429] if the next transmission occasion for the selected sidelink grant may satisfy the delay requirement or the remaining PDB of the MAC PDU (and the next transmission occasion is within the next COT):

[0430] The UE may perform retransmission of the MAC PDU using the next transmission occasion of the sidelink grant selected in the next COT.

[0431] Otherwise (i.e., the next transmission occasion for the selected sidelink grant cannot satisfy the delay requirement or the remaining PDB of the MAC PDU):

[0432] Alternative 1-1: The UE may reserve additional resource(s) within the COT and add the additional reserved resource(s) to the selected sidelink grant (i.e., the UE may not clear the selected sidelink grant).

[0433] Alternative 1-2: The UE may clear the selected sidelink grant associated with the sidelink process and trigger TX resource (re)selection.

[0434] If the UE is unable to reserve (additional) resource(s) within the COT according to Alternative 1 or Alternative 1-2, and the next transmission occasion of the selected sidelink grant cannot satisfy the delay requirement or the remaining PDB of the MAC PDU, the UE may drop the MAC PDU, flush the buffer of the HARQ process where the MAC PDU is stored, or consider that the UE has reached the maximum number of transmissions of the MAC PDU.

[0435] When the MAC entity performs TX resource (re) selection check for the resource pool selected for the corresponding sidelink process, the TX UE may release the selected sidelink grant associated with the sidelink process, if possible, and trigger TX resource (re)selection.

[0436] As an example of the present disclosure, the UE may perform at least one of the alternatives described below.

[0437] if a consistent LBT failure is triggered for the selected sidelink grant; or

[0438] if a consistent LBT failure is triggered for the active SL BWP of the corresponding serving cell associated with the selected sidelink grant according to the above-described method; or

[0439] if a consistent LBT failure is triggered for all SL BWPs of the corresponding serving cell associated with the selected sidelink grant; or

[0440] if a consistent LBT failure is triggered for SL or UL on the corresponding serving cell associated with the selected sidelink grant and is not cancelled:

[0441] Alternative 6-1: The UE may reselect from an old SL resource pool to a new resource pool that triggers reselection to generate a sidelink grant newly selected by the UE. Here, the currently selected sidelink grant may be reserved in the old SL resource pool.

[0442] Alternative 6-2: The UE may reselect from an old RB set of an activated SL BWP to a new RB set of an activated SL BWP that triggers reselection to generate a sidelink grant newly selected by the terminal. Here, the currently selected sidelink grant may be reserved in the old RB set.

[0443] Alternative 6-3: The UE may reselect from an old SL BWP to a new SL BWP that triggers reselection to generate a sidelink grant newly selected by the terminal. Here, the currently selected sidelink grant may be reserved in the old SL BWP set.

[0444] Alternative 6-4: The UE may reselect to a new cell, which triggers reselection to generate a newly selected sidelink grant from the previous cell. Here, the currently selected sidelink grant may be reserved on the carrier of the previous cell.

[0445] In the alternative described above, 'lbt-FailureInstanceMaxCount' for consistent LBT failure detection may be set to different values for different alternatives.

[0446] Additionally or alternatively, 'lbt-FailureDetectionTimer' for consistent LBT failure detection may be set to different values for different alternatives.

[0447] For example, the base station may set different 'lbt-FailureInstanceMaxCount' values for different alternatives. For example, the 'lbt-FailureInstanceMaxCount' value for Alternative 6-1 may be lower than the 'lbt-FailureInstanceMaxCount' value for Alternative 6-3.

[0448] Alternatively, the base station may set different 'lbt-FailureDetectionTimer' values for different alternatives such that the 'lbt-FailureDetectionTimer' value for Alternative 6-4 may be lower than the 'lbt-FailureDetectionTimer' value for Alternative 6-3.

[0449] 'lbt-FailureInstanceMaxCount' for consistent LBT failure detection and 'lbt-FailureDetectionTimer' for consistent LBT failure detection may be set to different values for SL mode 1 (i.e., SL resource allocation mode 1) and SL mode 2 (i.e., SL resource allocation mode 2), respectively.

[0450] For SL mode 1, the parameters described above may be used for SR reporting. For SL mode 2, the parameters described above may be used for resource reselection. Additionally or alternatively, SR may be triggered due to consistent LBT errors in the steps described above for both SL mode 1 and SL mode 2.

[0451] Additionally or alternatively, the UE may release the PC5-RRC connection based on consistent LBT failures as described below.

[0452] If a consistent LBT failure is triggered for the selected sidelink grant: or

[0453] if a consistent LBT failure is triggered for an active SL BWP of the corresponding serving cell associated with the selected sidelink grant in the step described above: or

[0454] if a consistent LBT failure is triggered for all SL BWPs of the corresponding serving cell associated with the selected sidelink grant: or

[0455] if a consistent LBT failure is triggered and not canceled for SL or UL, then on the corresponding serving cell associated with the selected sidelink grant:

[0456] If the TX UE establishes a PC5-RRC connection with the RX UE,

[0457] Option 6-1: The MAC entity of the TX UE may consider that a sidelink RLF has been detected and may notify the RRC of the TX UE of 'Sidelink RLF Detection'.

[0458] Option 6-2: The MAC entity of the TX UE may consider that a consistent SL LBT failure has been detected and may notify the RRC of the TX UE of 'Consistent SL LBT Failure'.

[0459] The TX UE may send a PC5-RRC release message to the RX UE and release the PC5-RRC connection with the RX UE. The RX UE that receives the PC5-RRC release message may release the PC5-RRC connection with the TX UE.

[0460] The TX UE may transmit a PUCCH to the base station to indicate persistent LBT failure or PC5-RRC disconnection by triggering SR.

[0461] After the PC5-RRC disconnection, the TX UE may transmit a 'SidelinkUEInformation' message to the base station to indicate the disconnection of the PC5-RRC with the RX UE. The 'SidelinkUEInformation' message may indicate to the base station that the PC5-RRC with the RX

terminal has been disconnected or should be disconnected due to consistent LBT failure detection.

[0462] The UE may perform re-evaluation or pre-emption according to the method described below.

[0463] Resource(s) of the sidelink grant selected for transmission from the multiplexing and assembling entity for MAC PDU may be reevaluated by the physical layer in the T3 slot prior to the slot in which the SCI indicating the resource(s) is first signaled. The resource(s) of the selected sidelink grant indicated by the previous SCI for transmission from the multiplexing and assembling entity for MAC PDU may be examined for preemption by the physical layer in the T3 slot prior to the slot in which the resource(s) are located.

[0464] Re-evaluation or preemption before 'm-T3' or after 'm-T3' but before 'm' may vary depending on the terminal implementation. For re-evaluation, m means the slot in which the SCI indicating the resource(s) is initially signaled. For preemption, m may mean the slot in which the resource is located.

[0465] When sidelink resource allocation mode 2 is set to transmit using the carrier's resource pool(s) for the MAC entity of the UE or to transmit based on detection or random selection, the MAC entity of the UE may perform the following for each sidelink process:

[0466] 1> if the resource(s) of the selected sidelink grant not identified by the previous SCI are indicated for re-evaluation by the physical layer;

[0467] 2> remove the resource(s) from the selected sidelink grant associated with the sidelink process;

[0468] 2> if the PSFCH is configured for the corresponding resource pool and the resource may be indicated by the time resource allocation of the SCI for retransmission, then randomly select the time and frequency resources from the resources indicated by the physical layer for the removed or dropped resource, by ensuring a minimum time interval between the selected two resources of the selected sidelink grant, based on the amount of the selected frequency resources, the number of selected HARQ retransmissions, and the remaining PDB of SL data available on the logical channel;

[0469] 2> replace the selected resource with the selected resource for the sidelink grant that selected the removed or deleted resource.

[0470] 1> if the resource(s) of the selected sidelink grant are not within the COT (Channel Occupancy Time) duration determined by the channel access procedure performed by the UE or shared by any UE/gNB/eNB; or

[0471] 1> if the resource(s) of the selected sidelink grant cannot be used for SL transmission due to LBT failure indication of L1;

[0472] 2> remove a resource from the selected sidelink grant associated with the sidelink process;

[0473] 2> Select time and frequency resources randomly from among the resources indicated by the physical layer for the removed or dropped resources, by ensuring a minimum time interval between any two selected resources of the selected sidelink grant, if PSFCH is configured for the corresponding resource pool and the resources may be indicated by the time resource allocation of SCI for retransmission, based on the amount of frequency resources randomly selected,

the number of selected HARQ retransmissions, and the remaining PDB of SL data available on the logical channel;

[0474] 2> replace the selected resource with the selected resource for the selected sidelink grant that has been removed or deleted.

[0475] 1> if the resource(s) of the selected sidelink grant indicated by the previous SCI are marked for preemption by the physical layer;

[0476] 2> if more than one SL DRX is configured;

[0477] 3> remove resource(s) from selected sidelink grant associated with sidelink process;

[0478] 3> if PSFCH is configured for the corresponding resource pool and the resource may be indicated by the time resource allocation of SCI for retransmission, by ensuring the minimum time interval between the selected two resources of the selected sidelink grant, the amount of frequency resources, the number of HARQ retransmissions selected and the remaining PDB of SL data available on the logical channel, randomly selecting time and frequency resources from later resources than the resources for the removed or missing resources indicated by the previous SCI;

[0479] 3> replace the selected resource with the selected resource for the sidelink grant that selected the removed or deleted resource.

[0480] 2> else;

[0481] 3> remove resource(s) from selected sidelink grant associated with sidelink process;

[0482] 3> If PSFCH is configured for the corresponding resource pool and the resource may be indicated by the time resource allocation of SCI for retransmission, the time and frequency resources are selected randomly from among the resources indicated by the physical layer for the removed or dropped resources by ensuring the minimum time interval between the selected two resources of the selected sidelink grant, the amount of frequency resources, the number of selected HARQ retransmissions and the remaining PDB of SL data available on the logical channel;

[0483] 3> replace the selected resource with the selected resource for the sidelink grant that selected the removed or deleted resource.

[0484] Here, if it is not possible to select the retransmission resource(s) by ensuring that the resource(s) may be indicated by the time resource allocation of the previous SCI, the method of selecting the time and frequency resources for one or more transmission occasions from the available resources may be based on the UE implementation by ensuring the minimum time interval between the two selected resources when the PSFCH is configured for that resource pool.

[0485] A UE implementation may be left to reselect a pre-selected but unreserved resource other than the resource indicated for preemption or re-evaluation by the physical layer during re-evaluation or preemption triggered by the physical layer.

[0486] Whether to set a resource reservation interval on a reselected resource to replace a preempted resource may vary by UE implementation. Whether to trigger resource reselection due to deprioritization may vary by UE implementation.

[0487] If the selected sidelink grant corresponds to the transmission of multiple MAC PDUs, it may be up to the UE

implementation whether re-evaluation checks are applied to resources of non-initial reservation periods that were not signaled in the last or current period.

[0488] At a TX UE, for each SCI and SL grant corresponding to a new transmission, the MAC entity may perform a logical channel priority procedure to select a destination associated with one of the unicast, groupcast and broadcast logical channels and MAC CEs having at least one of the logical channels and MAC CEs with the highest priority (for the SL grant associated with the SCI) that satisfy all of the following conditions:

[0489] Here, the conditions may include that i) SL data may be transmitted, ii) there is a logical channel with SB_j (sidelink bucket size/value in logical channel j) > 0 and $SB_j > 0$, iii) 'sl-configuredGrantType1Allowed' is set to 'true' if the SL grant is of configured grant type 1 (if set), iv) 'sl-AllowedCG-List' contains configured grant indices associated with the SL grant (if set), and v) 'sl-HARQ-FeedbackEnabled' is set to disabled if PSFCH is not configured for an SL grant associated with SCI.

[0490] If multiple destinations have logical channels satisfying all of the above conditions with the same highest priority, or if multiple destinations have logical channels satisfying all of the above conditions with the same priority as the MAC CE and/or the MAC CE, selecting a particular destination among the above-described destinations may be terminal implementation-based.

[0491] If the PSFCH is configured for a sidelink grant associated with SCI at that step and any consecutive SL grants used for transmission of MAC PDUs to be generated for that procedure can satisfy the minimum time interval:

[0492] The MAC entity sets 'sl-HARQ-FeedbackEnabled' to enabled if 'sl-HARQ-FeedbackEnabled' is set to enabled for the logical channel with the highest priority satisfying the above conditions: or

[0493] A MAC entity may set 'sl-HARQ-FeedbackEnabled' to disabled if 'sl-HARQ-FeedbackEnabled' is set to disabled for the logical channel with the highest priority satisfying the above conditions.

[0494] Meanwhile, if PSFCH is not configured for a sidelink grant related to SCI or if consecutive SL grants used for transmission of MAC PDUs generated for the procedure cannot meet the minimum time interval, the MAC entity may set 'sl-HARQ-FeedbackEnabled' to disabled.

[0495] When a MAC entity sets 'sl-HARQ-FeedbackEnabled' to enabled, the MAC entity may select a logical channel with 'sl-HARQ-FeedbackEnabled' set to enabled. When a MAC entity sets 'sl-HARQ-FeedbackEnabled' to disabled, the MAC entity can select a logical channel with 'sl-HARQ-FeedbackEnabled' set to disabled.

[0496] And, for each SCI and SL grant corresponding to a new transmission, the MAC entity can generate MAC PDU including SL data only on the logical channel(s) selected for the HARQ entity. Here, a logical channel with 'sl-HARQ-FeedbackEnabled' set to enable and a logical channel with 'sl-HARQ-FeedbackEnabled' set to disable may not be multiplexed in the same MAC PDU.

[0497] At the transmitting UE, if the MAC entity determines that the sidelink grant is used for the initial transmission for each sidelink grant of the selected sidelink grant, or the sidelink grant is a configured sidelink grant and no MAC PDU is acquired in the 'sl-PeriodCG' of the configured sidelink grant, the MAC entity may acquire the MAC PDU generated in the above-described step.

[0498] When a MAC PDU to be transmitted is acquired, the MAC entity may determine the sidelink transmission information of the TB for the source and destination pair of the MAC PDU as described below.

[0499] When a MAC PDU to be transmitted is acquired, the MAC entity may determine the sidelink transmission information of the TB for the source and destination pair of the MAC PDU as follows:

[0500] The MAC entity may configure the source layer-1 ID to the 8 LSB of the source layer-2 ID of the MAC PDU.

[0501] The MAC entity may configure the destination layer-1 ID to the 16 LSB of the destination layer-2 ID of the MAC PDU.

[0502] The MAC entity may consider the NDI to be toggled by comparing the sidelink identification information with the previous transmission value corresponding to the sidelink process ID of the MAC PDU and may set the NDI to the toggled value.

[0503] The MAC entity may configure the cast type indicator to one of broadcast, groupcast, and unicast as directed by the upper layer.

[0504] If HARQ feedback is enabled for the MAC PDU according to the logical channel priority procedure, the MAC entity may set the HARQ feedback enable/disable indicator to enabled.

[0505] If HARQ feedback is disabled for the MAC PDU according to the logical channel priority procedure, the MAC entity may set the HARQ feedback enable/disable indicator to disabled.

[0506] The MAC entity may configure the priority to the highest priority value of the logical channel (if any) and the MAC CE (if included) included in the MAC PDU.

[0507] If both the group size and the member ID are provided by upper layers and the group size is not greater than the number of candidate PSFCH resources associated with the corresponding sidelink grant, the MAC entity may select a positive-negative ACK or a negative-only grant. Otherwise, the MAC entity may select a negative-only grant.

[0508] If negative-only ACK is selected, the location information of the UE is available, 'sl-TransRange' is set for the logical channel in the MAC PDU, and 'sl-ZoneConfig' is set, the MAC entity may configure the communication range requirement to the longest communication range value of the logical channel in the MAC PDU.

[0509] The MAC entity may determine the 'sl-Zone-Length' value corresponding to the communication range requirement, set the 'Zone_id' to the 'Zone_id' value calculated using the determined 'sl-ZoneLength' value, and perform a new transmission of the MAC PDU on the PSSCH using the sidelink grant.

[0510] The TX UE may monitor the PSFCH occasion and determine positive ACK or negative ACK for the MAC PDU based on the PSFCH transmission performed by the RX UE.

[0511] If the TX UE determines negative ACK and the sidelink grant of the selected sidelink grant is retransmittable, the MAC entity may perform retransmission of the MAC PDU over the PSSCH using the sidelink grant.

[0512] In describing the present disclosure, sidelink transmission, grant and resource may be replaced with uplink transmission, grant and resource. (When the UE autonomously

mously determines a new transmission or retransmission up to a maximum of N transmissions of TB in the resource pool) the UE may change the resource pool or SL BWP or RB set or serving cell in which the SL transmission is performed based on consistent SL LBT failure.

General Device to which the Present Disclosure May be Applied

[0513] FIG. 13 illustrates a block diagram of a wireless communication device according to an embodiment of the present disclosure.

[0514] In reference to FIG. 13, a first wireless device 100 and a second wireless device 200 may transmit and receive a wireless signal through a variety of radio access technologies (e.g., LTE, NR).

[0515] A first wireless device 100 may include one or more processors 102 and one or more memories 104 and may additionally include one or more transceivers 106 and/or one or more antennas 108. A processor 102 may control a memory 104 and/or a transceiver 106 and may be configured to implement description, functions, procedures, proposals, methods and/or operation flow charts disclosed in the present disclosure.

[0516] For example, the processor 102 may process the information in the memory 104 to generate first information/signal and then transmit a wireless signal including the first information/signal through the transceiver 106. Additionally, the processor 102 may receive a wireless signal including the second information/signal through the transceiver 106 and then store information obtained from signal processing of the second information/signal in the memory 104.

[0517] A memory 104 may be connected to a processor 102 and may store a variety of information related to an operation of a processor 102. For example, a memory 104 may store a software code including commands for performing all or part of processes controlled by a processor 102 or for performing description, functions, procedures, proposals, methods and/or operation flow charts disclosed in the present disclosure. Here, a processor 102 and a memory 104 may be part of a communication modem/circuit/chip designed to implement a wireless communication technology (e.g., LTE, NR). A transceiver 106 may be connected to a processor 102 and may transmit and/or receive a wireless signal through one or more antennas 108. A transceiver 106 may include a transmitter and/or a receiver. A transceiver 106 may be used together with a RF (Radio Frequency) unit. In the present disclosure, a wireless device may mean a communication modem/circuit/chip.

[0518] A second wireless device 200 may include one or more processors 202 and one or more memories 204 and may additionally include one or more transceivers 206 and/or one or more antennas 208. A processor 202 may control a memory 204 and/or a transceiver 206 and may be configured to implement description, functions, procedures, proposals, methods and/or operation flows charts disclosed in the present disclosure. For example, a processor 202 may generate third information/signal by processing information in a memory 204, and then transmit a wireless signal including third information/signal through a transceiver 206. In addition, a processor 202 may receive a wireless signal including fourth information/signal through a transceiver 206, and then store information obtained by signal processing of fourth information/signal in a memory 204. A memory 204 may be connected to a processor 202 and may store a variety of information related to an operation of a processor

202. For example, a memory 204 may store a software code including commands for performing all or part of processes controlled by a processor 202 or for performing description, functions, procedures, proposals, methods and/or operation flow charts disclosed in the present disclosure. Here, a processor 202 and a memory 204 may be part of a communication modem/circuit/chip designed to implement a wireless communication technology (e.g., LTE, NR). A transceiver 206 may be connected to a processor 202 and may transmit and/or receive a wireless signal through one or more antennas 208. A transceiver 206 may include a transmitter and/or a receiver. A transceiver 206 may be used together with a RF unit. In the present disclosure, a wireless device may mean a communication modem/circuit/chip.

[0519] Hereinafter, a hardware element of a wireless device 100, 200 will be described in more detail. It is not limited thereto, but one or more protocol layers may be implemented by one or more processors 102, 202. For example, one or more processors 102, 202 may implement one or more layers (e.g., a functional layer such as PHY, MAC, RLC, PDCP, RRC, SDAP). One or more processors 102, 202 may generate one or more PDUs (Protocol Data Unit) and/or one or more SDUs (Service Data Unit) according to description, functions, procedures, proposals, methods and/or operation flow charts included in the present disclosure. One or more processors 102, 202 may generate a message, control information, data or information according to description, functions, procedures, proposals, methods and/or operation flow charts disclosed in the present disclosure. One or more processors 102, 202 may generate a signal (e.g., a baseband signal) including a PDU, a SDU, a message, control information, data or information according to functions, procedures, proposals and/or methods disclosed in the present disclosure to provide it to one or more transceivers 106, 206. One or more processors 102, 202 may receive a signal (e.g., a baseband signal) from one or more transceivers 106, 206 and obtain a PDU, a SDU, a message, control information, data or information according to description, functions, procedures, proposals, methods and/or operation flow charts disclosed in the present disclosure.

[0520] One or more processors 102, 202 may be referred to as a controller, a micro controller, a micro processor or a micro computer. One or more processors 102, 202 may be implemented by a hardware, a firmware, a software, or their combination. In an example, one or more ASICs (Application Specific Integrated Circuit), one or more DSPs (Digital Signal Processor), one or more DSPDs (Digital Signal Processing Device), one or more PLDs (Programmable Logic Device) or one or more FPGAs (Field Programmable Gate Arrays) may be included in one or more processors 102, 202. Description, functions, procedures, proposals, methods and/or operation flow charts disclosed in the present disclosure may be implemented by using a firmware or a software and a firmware or a software may be implemented to include a module, a procedure, a function, etc. A firmware or a software configured to perform description, functions, procedures, proposals, methods and/or operation flow charts disclosed in the present disclosure may be included in one or more processors 102, 202 or may be stored in one or more memories 104, 204 and driven by one or more processors 102, 202. Description, functions, procedures, proposals, methods and/or operation flow charts disclosed in the pres-

ent disclosure may be implemented by using a firmware or a software in a form of a code, a command and/or a set of commands.

[0521] One or more memories **104, 204** may be connected to one or more processors **102, 202** and may store data, a signal, a message, information, a program, a code, an instruction and/or a command in various forms. One or more memories **104, 204** may be configured with ROM, RAM, EPROM, a flash memory, a hard drive, a register, a cash memory, a computer readable storage medium and/or their combination. One or more memories **104, 204** may be positioned inside and/or outside one or more processors **102, 202**. In addition, one or more memories **104, 204** may be connected to one or more processors **102, 202** through a variety of technologies such as a wire or wireless connection.

[0522] One or more transceivers **106, 206** may transmit user data, control information, a wireless signal/channel, etc. mentioned in methods and/or operation flow charts, etc. of the present disclosure to one or more other devices. One or more transceivers **106, 206** may receive user data, control information, a wireless signal/channel, etc. mentioned in description, functions, procedures, proposals, methods and/or operation flow charts, etc. disclosed in the present disclosure from one or more other devices. For example, one or more transceivers **106, 206** may be connected to one or more processors **102, 202** and may transmit and receive a wireless signal. For example, one or more processors **102, 202** may control one or more transceivers **106, 206** to transmit user data, control information or a wireless signal to one or more other devices. In addition, one or more processors **102, 202** may control one or more transceivers **106, 206** to receive user data, control information or a wireless signal from one or more other devices. In addition, one or more transceivers **106, 206** may be connected to one or more antennas **108, 208** and one or more transceivers **106, 206** may be configured to transmit and receive user data, control information, a wireless signal/channel, etc. mentioned in description, functions, procedures, proposals, methods and/or operation flow charts, etc. disclosed in the present disclosure through one or more antennas **108, 208**. In the present disclosure, one or more antennas may be a plurality of physical antennas or a plurality of logical antennas (e.g., an antenna port). One or more transceivers **106, 206** may convert a received wireless signal/channel, etc. into a baseband signal from a RF band signal to process received user data, control information, wireless signal/channel, etc. by using one or more processors **102, 202**. One or more transceivers **106, 206** may convert user data, control information, a wireless signal/channel, etc. which are processed by using one or more processors **102, 202** from a baseband signal to a RF band signal. Therefore, one or more transceivers **106, 206** may include an (analogue) oscillator and/or a filter.

[0523] Embodiments described above are that elements and features of the present disclosure are combined in a predetermined form. Each element or feature should be considered to be optional unless otherwise explicitly mentioned. Each element or feature may be implemented in a form that it is not combined with other element or feature. In addition, an embodiment of the present disclosure may include combining a part of elements and/or features. An order of operations described in embodiments of the present disclosure may be changed. Some elements or features of one embodiment may be included in other embodiment or

may be substituted with a corresponding element or a feature of other embodiment. It is clear that an embodiment may include combining claims without an explicit dependency relationship in claims or may be included as a new claim by amendment after application.

[0524] It is clear to a person skilled in the pertinent art that the present disclosure may be implemented in other specific form in a scope not going beyond an essential feature of the present disclosure. Accordingly, the above-described detailed description should not be restrictively construed in every aspect and should be considered to be illustrative. A scope of the present disclosure should be determined by reasonable construction of an attached claim and all changes within an equivalent scope of the present disclosure are included in a scope of the present disclosure.

[0525] A scope of the present disclosure includes software or machine-executable commands (e.g., an operating system, an application, a firmware, a program, etc.) which execute an operation according to a method of various embodiments in a device or a computer and a non-transitory computer-readable medium that such a software or a command, etc. are stored and are executable in a device or a computer. A command which may be used to program a processing system performing a feature described in the present disclosure may be stored in a storage medium or a computer-readable storage medium and a feature described in the present disclosure may be implemented by using a computer program product including such a storage medium. A storage medium may include a high-speed random-access memory such as DRAM, SRAM, DDR RAM or other random-access solid state memory device, but it is not limited thereto, and it may include a nonvolatile memory such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices or other non-volatile solid state storage devices. A memory optionally includes one or more storage devices positioned remotely from processor(s). A memory or alternatively, nonvolatile memory device(s) in a memory include a non-transitory computer-readable storage medium. A feature described in the present disclosure may be stored in any one of machine-readable mediums to control a hardware of a processing system and may be integrated into a software and/or a firmware which allows a processing system to interact with other mechanism utilizing a result from an embodiment of the present disclosure. Such a software or a firmware may include an application code, a device driver, an operating system and an execution environment/container, but it is not limited thereto.

[0526] Here, a wireless communication technology implemented in a wireless device **100, 200** of the present disclosure may include Narrowband Internet of Things for a low-power communication as well as LTE, NR and 6G. Here, for example, an NB-IoT technology may be an example of a LPWAN (Low Power Wide Area Network) technology, may be implemented in a standard of LTE Cat NB1 and/or LTE Cat NB2, etc. and is not limited to the above-described name. Additionally or alternatively, a wireless communication technology implemented in a wireless device **100, 200** of the present disclosure may perform a communication based on a LTE-M technology. Here, in an example, a LTE-M technology may be an example of a LPWAN technology and may be referred to a variety of names such as an eMTC (enhanced Machine Type Communication), etc. For example, an LTE-M technology may be

implemented in at least any one of various standards including 1) LTE CAT 0, 2) LTE Cat M1, 3) LTE Cat M2, 4) LTE non-BL (non-Bandwidth Limited), 5) LTE-MTC, 6) LTE Machine Type Communication, and/or 7) LTE M and so on and it is not limited to the above-described name. Additionally or alternatively, a wireless communication technology implemented in a wireless device **100**, **200** of the present disclosure may include at least any one of a ZigBee, a Bluetooth and a low power wide area network (LPWAN) considering a low-power communication and it is not limited to the above-described name. In an example, a ZigBee technology may generate PAN (personal area networks) related to a small/low-power digital communication based on a variety of standards such as IEEE 802.15.4, etc. and may be referred to as a variety of names.

[0527] A method proposed by the present disclosure is mainly described based on an example applied to 3GPP LTE/LTE-A, 5G system, but may be applied to various wireless communication systems other than the 3GPP LTE/LTE-A, 5G system.

1-15. (canceled)

16. A method performed by a first user equipment (UE) in a wireless communication system, the UE comprising:

detecting a sidelink (SL) radio link failure of a second UE based on at least one indication of consistent SL listen before talk (LBT) failure for all resource block (RB) sets for the second UE; and

transmitting, to a base station, sidelink UE information including information related to the SL radio link failure,

wherein a PC5-radio resource control (RRC) connection for the second UE is released, and

wherein the PC5-RRC connection release is indicated to a upper layer for the second UE.

17. The method of claim **16**, wherein:

a higher layer signaling including information for releasing the PC5-RRC connection of the second UE related to the consistent SL LBT failure is transmitted by the first UE to the second UE.

18. The method of claim **16**, wherein:

first configuration information related to a SL LBT failure detection timer and second configuration information related to a maximum number of SL LBT failure indications for the consistent SL LBT failure are received from the base station.

19. The method of claim **18**, wherein:

the SL LBT failure detection timer and the maximum number of SL LBT failure for the consistent SL LBT failure are configured for the first UE on SL bandwidth part (BWP).

20. The method of claim **16**, wherein:

information indicating at least one of the consistent SL-specific LBT failure or release of the PC5-RRC connection is transmitted to the base station via a physical uplink control channel (PUCCH) by triggering a scheduling request (SR).

21. The method of claim **18**, wherein:

the first configuration information and the second configuration information are received from the base station through a higher layer signaling related to SL-specific LBT failure recovery.

22. The method of claim **18**, wherein:

the consistent SL LBT failure is triggered based on the SL LBT failure detection timer and the maximum number of SL LBT failure for the consistent SL LBT failure for at least one RB set.

23. The method of claim **18**, wherein:

based on at least one of the SL LBT failure detection timer and the maximum number of SL LBT failure for the consistent SL LBT failure being reconfigured for the UE, a value of a SL LBT counter is set to 0.

24. The method of claim **18**, wherein:

based on an expiration of the SL LBT failure detection timer, a value of a SL LBT counter is set to 0.

25. A first user equipment (UE) in a wireless communication system, the first UE comprising:

at least one transceiver; and

at least one processor connected to the at least one transceiver,

wherein the at least one processor is configured to:

detect a sidelink (SL) radio link failure of a second UE based on at least one indication of consistent SL listen before talk (LBT) failure for all resource block (RB) sets for the second UE;

transmit, to the base station through the at least one transceiver, sidelink UE information including information related to the SL radio link failure,

wherein a PC5-radio resource control (RRC) connection for the second UE is released, and

wherein the PC5-RRC connection release is indicated to a upper layer for the second UE.

* * * * *